



On a gamma–beta prime mixture model for heterogeneous positive data: properties and inference

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Abstract

In this paper, we introduce mixture distribution of gamma and beta prime. The gamma distribution is highly versatile for modeling continuous, positive-valued data, especially when the data exhibits skewness or variability in scale. The beta prime distribution also known as the beta distribution of the second kind, extends the beta distribution to support modeling data with heavy tails and more complex shapes. Combining these two distributions into a mixture allows the model to adapt multi modal features or heterogeneity that cannot be captured by a single distribution. This model can handle data that comes from different sources or behaves in different ways, making it more accurate and easier to understand. Some important statistical and mathematical properties of the distribution are derived such as the distribution function, moment, skewness, kurtosis, and the order statistics. The maximum likelihood estimation method is adopted to estimate the model parameters and a simulation study is conducted to investigate the efficiency of the various estimators. To evaluate the characteristics and inequality properties of the proposed distribution, we compute the Renyi and Tsallis entropy measures, Lorenz and Bonferroni curves, and the Gini index.

Keywords Mixture distribution · Moments · Tail decay · Order statistics · Simulation · Applications

1 Introduction

The main aim of probability distributions is to create statistically robust models that offer mathematical depictions of real-world data. Consequently, choosing the right distribution is vital for making dependable inferences and accurate forecasts. In many cases, real-world datasets display significant positive skewness, with data points clustered towards the higher end of the distribution. Traditional models like the exponential, Weibull, Rayleigh, and gamma distributions are frequently used to analyze this kind of data. Nonetheless, these models have inherent drawbacks, as their hazard rate structures are generally limited to being either constant or monotonically increasing, which restricts their capacity to capture the intricate data patterns encountered in actual applications. To address these limitations,

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several distributional generalization techniques have been proposed, including weighted (Ranade and Rather 2024), transmuted (Ali et al. 2024; Qayoom and Rather 2024), exponentiated (Nadarajah and Gupta 2007; Singh et al. 2023), mixture (McNicholas 2016; Ibrahim et al. 2026; Cai et al. 2026; Salah et al. 2025; Elgarhy et al. 2024) and compound distributions (Willmot 1986). These approaches introduce additional flexibility in shape, tail behavior, and hazard structure, leading to improved modeling performance for heterogeneous datasets.

Among continuous distributions, the Gamma distribution plays a central role in reliability engineering, actuarial science, and Bayesian analysis due to its mathematical tractability and flexibility for modeling positive-valued data. Nevertheless, the classical Gamma model may be inadequate for data exhibiting pronounced skewness or heavy tails. Consequently, numerous extensions of the Gamma distribution have been developed to enhance tail behavior and shape flexibility. Notable examples include the weighted Gamma distribution (Dennis and Patil 1984), the exponentiated Gamma distribution (Nadarajah and Gupta 2007), the exponentiated generalized Gamma distribution (Cordeiro et al. 2011), and the transmuted Gamma distribution (Lucena et al. 2015). More recently, poly-weighted and interpretable Gamma-type models have been proposed to further improve flexibility and parameter interpretability (Nkemnole and Ikegwu 2020; Wagener et al. 2024).

Gamma-based models have also been integrated with extreme value theory to better capture tail risk. For instance, Zheng et al. (2014) proposed a shifted Gamma–generalized Pareto model for traffic safety analysis, demonstrating improved performance over classical approaches. Theoretical investigations have further established the relevance of generalized Gamma distributions in extreme value analysis and reliability theory (Du and Chen 2016).

Parallel to Gamma-type developments, the Beta and Beta-prime distributions have received considerable attention for modeling bounded and heavy-tailed data. Extensions of the Beta family have been explored to enhance shape control and adaptability (McDonald and Xu 1995; Mauldon 1959), while the Beta-prime distribution has proven particularly useful in finance and reliability applications due to its heavy-tailed nature.

The complementary flexibility of the Gamma and Beta-prime distributions motivates their combination within mixture modeling frameworks. Mixture models are well known for their ability to capture multimodality, skewness, and tail heterogeneity (Willmot 1986). Several studies have demonstrated the effectiveness of Gamma and Beta-prime mixtures in modeling complex data structures (Wiper et al. 2001; Ji et al. 2005). These observations motivate the development of new mixture distributions that leverage the strengths of both components to achieve improved flexibility and robustness.

This research investigates the behavior and utility of mixture modelling and will help in contributing to a growing body of knowledge through an extensive literature review of advanced distribution modelling techniques. This research will consider the Gamma and Beta prime distributions due to the difference in tail characteristics of these two distributions where the Gamma Distribution adequately fits light to moderate tail behaviour and the Beta prime distribution can provide more flexibility to fit heavy-tailed behaviours of positive-valued data.

Motivated by parsimony and interpretability, we propose a mixture framework in which the same scale parameter β is used for both the Gamma and Beta prime components. This assumption reduces model complexity by limiting the number of parameters to be estimated and facilitates interpretation by enforcing a common scale across components. Moreover,

sharing the scale parameter improves numerical stability in parameter estimation, particularly for small and moderate sample sizes. For the proposed mixture model, we derive several fundamental statistical properties, including the probability density function, cumulative distribution function, mean deviation, moments, mean, variance, survival function, and hazard rate.

2 Theoretical model

A mixture model is a probabilistic model in which the distribution of a random variable is expressed as a convex combination of two or more component distributions. If x is a random variable arising from a mixture of k distributions, then its probability density function (pdf) is given by

$$f(x) = \sum_{i=1}^k \pi_i f_i(x), \quad 0 \leq \pi_i \leq 1$$

where $f_i(x)$ denotes the probability density function of the i -th component distribution, π_i represents the mixing weight associated with the i -th component, satisfying $\sum_{i=1}^k \pi_i = 1$ (Mood et al. 1974).

Each observation is generated by first selecting a component distribution according to the mixing probabilities and then drawing a value from the selected distribution. In this paper, we consider a two-component additive finite mixture model consisting of the Gamma distribution and the Beta prime distribution. Throughout the paper, the parameter β is treated as a common parameter for both components of the mixture model. Let Y follow a Gamma distribution with parameters (α, β) . Its probability density function (pdf) is given by

$$f(y) = \frac{y^{\alpha-1} e^{-y/\beta}}{\Gamma(\alpha) \beta^\alpha}, \quad y > 0, \alpha > 0, \beta > 0. \quad (1)$$

The cumulative distribution function (cdf) of the Gamma distribution is

$$F(y) = \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)}, \quad (2)$$

where $\gamma\left(\alpha, \frac{y}{\beta}\right)$ denotes the lower incomplete gamma function.

If Y follows a Beta prime distribution (also known as the Beta distribution of the second kind) with parameters $\theta > 0$ and $\beta > 0$, then its pdf and cdf are given by

$$f(y) = \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)}, \quad y > 0, \quad (3)$$

where

$$B(\theta, \beta) = \frac{\Gamma(\theta)\Gamma(\beta)}{\Gamma(\theta + \beta)}.$$

The corresponding cdf is

$$F(y; \theta, \beta) = I_{\frac{y}{1+y}}(\theta, \beta), \quad (4)$$

where $I_x(\theta, \beta)$ denotes the regularized incomplete beta function.

Using Eqs. (1) and (3), we construct a mixture distribution with weights w and $1 - w$ as

$$f(y) = w f_{\text{Gamma}}(y; \alpha, \beta) + (1 - w) f_{\text{BetaPrime}}(y; \theta, \beta).$$

The mixing weights are defined as

$$w = \frac{\alpha\beta}{\alpha\beta + 1}, \quad 1 - w = \frac{1}{\alpha\beta + 1}.$$

Thus, pdf becomes

$$f(y) = \frac{\alpha\beta}{\alpha\beta + 1} \frac{y^{\alpha-1} e^{-y/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta + 1} \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)}. \quad (5)$$

Equation (5) is referred to as the pdf of mixture of gamma and beta prime distribution (GBPD). Similarly we can derive the cdf of GBPD as

$$\begin{aligned} F(y) &= w F_{\text{Gamma}}(y; \alpha, \beta) + (1 - w) F_{\text{BetaPrime}}(y; \theta, \beta) \\ F(y) &= \frac{\alpha\beta}{\alpha\beta + 1} F_{\text{Gamma}}(y; \alpha, \beta) + \frac{1}{\alpha\beta + 1} F_{\text{BetaPrime}}(y; \theta, \beta) \\ F(y) &= \frac{\alpha\beta}{\alpha\beta + 1} \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)} + \frac{1}{\alpha\beta + 1} I_{\frac{y}{1+y}}(\theta, \beta). \end{aligned} \quad (6)$$

Figure 1 indicates pdf plot of GBPD distribution and Fig. 2 indicates cdf plot of GBPD distribution for various parametrs.

3 Mean deviations

The mean deviation (MD) is a measure of dispersion that quantifies the average absolute deviation of a random variable from its mean. For the proposed mixture random variable Y with probability density function $f_Y(y)$ and mean $\mu = E(Y)$, the mean deviation is defined as:

$$\text{MD} = E[|Y - \mu|] = \int_0^\infty |y - \mu| f_Y(y) dy. \quad (7)$$

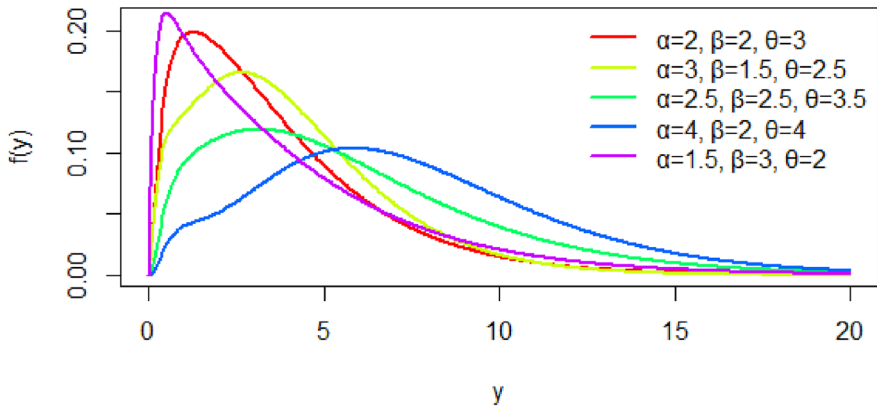


Fig. 1 pdf plot of GBPD

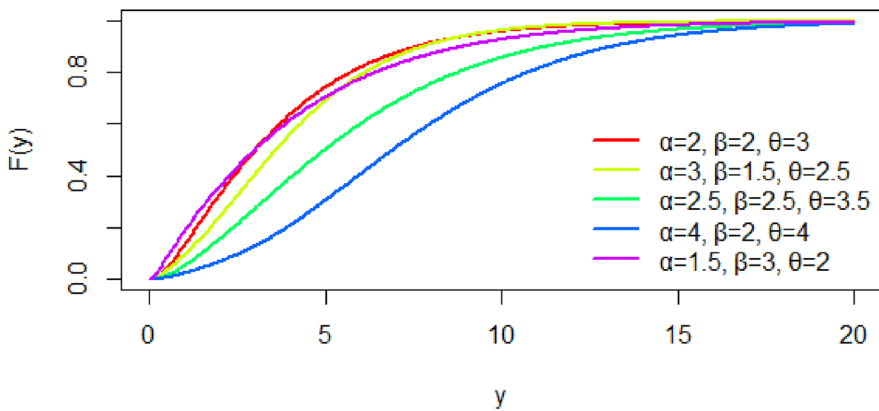


Fig. 2 cdf plot of GBPD

It can be expressed equivalently as

$$\text{MD} = \int_0^{\mu} (\mu - y) f_Y(y) dy + \int_{\mu}^{\infty} (y - \mu) f_Y(y) dy. \quad (8)$$

3.1 Mean deviation for the mixture of GBPD

Let

$$f(y) = w f_1(y; \alpha, \beta) + (1 - w) f_2(y; \theta, \beta), \quad (9)$$

where

$$w = \frac{\alpha\beta}{\alpha\beta + 1}. \quad (10)$$

Here, f_1 and f_2 denote the Gamma(α, β) and Beta-prime(θ, β) densities, respectively.

The mean of the mixture distribution is

$$\mu = \int_0^\infty y, f(y), dy. \quad (11)$$

The mean of the Beta-prime component exists only for ($\beta > 1$), ensuring the finiteness of μ . Substituting the mixture density given in Eq. (9) into the general expression for the mean deviation yields

$$\int_0^\mu (\mu - y)f(y), dy + \int_\mu^\infty (y - \mu)f(y), dy. \quad (12)$$

Since

$$|y - \mu| = \begin{cases} \mu - y, & y < \mu, \\ y - \mu, & y \geq \mu, \end{cases}$$

Now, expanding both integrals, we obtain

$$\text{MD} = \mu \int_0^\mu f(y) dy - \int_0^\mu yf(y) dy + \int_\mu^\infty yf(y) dy - \mu \int_\mu^\infty f(y) dy. \quad (13)$$

Using the identities

$$\int_0^\infty f(y) dy = 1, \quad \int_0^\infty yf(y) dy = \mu,$$

we write

$$\int_0^\mu yf(y) dy = \mu - \int_\mu^\infty yf(y) dy, \quad \int_0^\mu f(y) dy = 1 - \int_\mu^\infty f(y) dy.$$

Using Eq. (13), the mean deviation of the proposed GBPD mixture distribution can be obtained as follows.

$$\text{MD} = \mu - \mu - 2\mu \int_\mu^\infty f(y) dy + 2 \int_\mu^\infty yf(y) dy. \quad (14)$$

Hence, simplifying, we obtain

$$\text{MD} = 2 \int_\mu^\infty (y - \mu)f(y) dy. \quad (15)$$

By linearity of the integral and since the mixture mean μ is fixed, the mean deviation can be expressed as the weighted sum of the component contributions

$$MD = w MD_1 + (1 - w) MD_2 \quad (16)$$

where

$$MD_i = 2 \int_{\mu}^{\infty} (y - \mu) f_i(y) dy, \quad i = 1, 2.$$

For the Gamma component For $f_1(y)$ (Gamma(α, β)), one obtains

$$MD_1 = 2 \left[\int_{\mu}^{\infty} y f_1(y) dy - \mu (1 - F_1(\mu)) \right] \quad (17)$$

$$= 2 \left[\frac{\beta \Gamma(\alpha + 1, \frac{\mu}{\beta})}{\Gamma(\alpha)} - \mu \left(1 - \frac{\gamma(\alpha, \frac{\mu}{\beta})}{\Gamma(\alpha)} \right) \right], \quad (18)$$

where $\gamma(s, x)$ and $\Gamma(s, x)$ denote the lower and upper incomplete gamma functions, respectively, defined by

Beta-prime component Set $u = \frac{y}{1 + y}$, so $u \in (0, 1)$ and $u_0 = \frac{\mu}{1 + \mu}$. Then

$$\int_{\mu}^{\infty} y f_2(y) dy = \frac{1}{B(\theta, \beta)} \int_{u_0}^1 u^{\theta} (1 - u)^{\beta-2} du. \quad (19)$$

Hence, using the regularized incomplete beta function

$$I_x(a, b) = \frac{1}{B(a, b)} \int_0^x t^{a-1} (1 - t)^{b-1} dt$$

we obtain

$$MD_2 = 2 \left[\frac{1}{B(\theta, \beta)} \int_{u_0}^1 u^{\theta} (1 - u)^{\beta-2} du - \mu (1 - F_2(\mu)) \right] \quad (20)$$

$$= 2 \left[\frac{B(\theta + 1, \beta - 1)}{B(\theta, \beta)} (1 - I_{u_0}(\theta + 1, \beta - 1)) - \mu (1 - I_{u_0}(\theta, \beta)) \right]. \quad (21)$$

Noting the Beta identity

$$\frac{B(\theta + 1, \beta - 1)}{B(\theta, \beta)} = \frac{\theta}{\beta - 1}, \quad (22)$$

a convenient final form is

$$\text{MD}_2 = 2 \left[\frac{\theta}{\beta - 1} (1 - I_{u_0}(\theta + 1, \beta - 1)) - \mu (1 - I_{u_0}(\theta, \beta)) \right], \quad u_0 = \frac{\mu}{1 + \mu}. \quad (23)$$

Final mixture expression Combining the Eqs. (18) and (23), the MD of the mixture can be written compactly as

$$\begin{aligned} \text{MD} = & 2w \left[\frac{\beta \Gamma(\alpha + 1, \frac{\mu}{\beta})}{\Gamma(\alpha)} - \mu \left(1 - \frac{\gamma(\alpha, \frac{\mu}{\beta})}{\Gamma(\alpha)} \right) \right] \\ & + 2(1 - w) \left[\frac{\theta}{\beta - 1} (1 - I_{u_0}(\theta + 1, \beta - 1)) - \mu (1 - I_{u_0}(\theta, \beta)) \right]. \end{aligned} \quad (24)$$

Table 1 shows the comparison of mean deviation computed for the simulated data. The proposed model attains a mean deviation of (3.0867), which is very close to the weibull distribution (3.0703) and slightly higher than the gamma model (2.9993), indicating strong control over dispersion. In contrast, the lognormal and normal distributions exhibit higher variability, while the beta prime distribution shows extremely large dispersion (6.0299). This demonstrates that the proposed mixture distribution successfully balances tail flexibility with stability, making it highly suitable for inequality and risk modeling. The proposed model exhibits competitive and often reduced dispersion, confirming its suitability for inequality and risk assessment.

4 Gini index

The mean deviation provides a scalar measure of inequality and is closely related to the Lorenz and Bonferroni curves. Since these measures are constructed relative to the mean, the mean deviation offers a consistent assessment of dispersion and inequality. Moreover, the Gini coefficient can be expressed in terms of the mean deviation about the mean as

$$G = \frac{\text{MD}}{2\mu},$$

Substituting the closed-form expression of mean deviation derived in Eq. (24), we obtain

Table 1 Mean deviation (MD) for different distributions

Distribution	Mean deviation
GBPD	3.086725
Gamma	2.999292
Weibull	3.070294
Lognormal	3.938200
Normal	3.227712
Beta-prime	6.029877

$$G = \frac{1}{\mu} \left[w \left\{ \frac{\beta \Gamma\left(\alpha + 1, \frac{\mu}{\beta}\right)}{\Gamma(\alpha)} - \mu \left(1 - \frac{\gamma\left(\alpha, \frac{\mu}{\beta}\right)}{\Gamma(\alpha)} \right) \right\} + (1 - w) \left\{ \frac{\theta}{\beta - 1} (1 - I_{u_0}(\theta + 1, \beta - 1)) - \mu (1 - I_{u_0}(\theta, \beta)) \right\} \right]. \quad (25)$$

Thus, the Gini index is obtained in closed form in terms of the incomplete gamma and regularized incomplete beta functions. The Bonferroni curve is defined as The Bonferroni curve is defined as

$$B(p) = \frac{L(p)}{p}, \quad 0 < p \leq 1,$$

where $L(p)$ denotes the Lorenz curve. For the proposed distribution,

$$L(p) = \frac{1}{\mu} \int_0^{Q(p)} x f(x) dx, \quad 0 < p < 1,$$

where $Q(p) = F^{-1}(p)$ is the quantile function. Hence,

$$B(p) = \frac{1}{p\mu} \int_0^{Q(p)} x f(x) dx.$$

Substituting the proposed density function into the above integral yields the Bonferroni curve of the model. Owing to the complexity of the quantile function and the resulting integral, a simple closed-form expression for the Bonferroni curve is not available. Therefore, the Bonferroni curve may be evaluated numerically.

Table 2 compares the Gini index values for the competing distributions. The Proposed model yields a gini index of (0.3957460), which is very close to those of the gamma (0.3919541) and weibull (0.3864783) distributions, indicating that the proposed model captures a moderate level of dispersion while maintaining stability. Compared to the normal distribution (0.4182333), the proposed model exhibits slightly lower inequality, suggesting improved robustness against excessive variability while still accommodating skewness and tail behavior. Figures 3 and 4 illustrates the lorenz and bonferroni curves for the different distributions, highlighting the inequality patterns and differences in distributional behavior.

Table 2 Gini index for different distributions

Distribution	Gini index
GBPD	0.3957460
Gamma	0.3919541
Weibull	0.3864783
Lognormal	0.4684482
Normal	0.4182333
Beta-prime	0.5766471

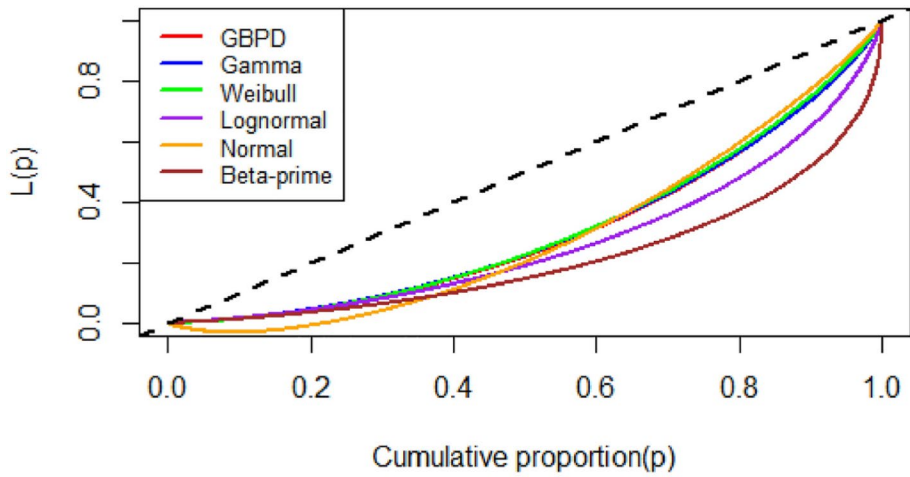


Fig. 3 Lorenz curves for various distributions

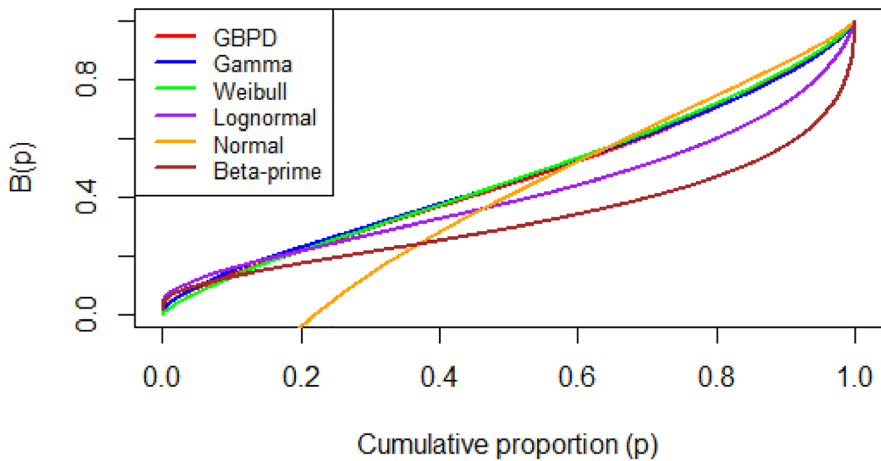


Fig. 4 Bonferroni curves for various distributions

5 Moments, skewness and kurtosis

Consider the mixture

$$Y \sim w Y_1 + (1 - w) Y_2, \quad w = \frac{\alpha\beta}{\alpha\beta + 1}, \quad (26)$$

where

$$Y_1 \sim \text{Gamma}(\alpha, \beta), \quad Y_2 \sim \text{Beta-prime}(\theta, \beta),$$

with parameters $\alpha, \beta, \theta > 0$.

Raw moments The general raw moment is

$$\mu'_r = E[Y^r] = w E_{Y_1}[Y^r] + (1 - w) E_{Y_2}[Y^r]. \quad (27)$$

For the component moments we have

$$E_{Y_1}[Y^r] = \beta^r \frac{\Gamma(\alpha + r)}{\Gamma(\alpha)}, \quad E_{Y_2}[Y^r] = \frac{B(\theta + r, \beta - r)}{B(\theta, \beta)} = \frac{\Gamma(\theta + r)\Gamma(\beta - r)}{\Gamma(\theta)\Gamma(\beta)},$$

with the Beta-prime r -th moment existing only if $\beta > r$.

Thus, for $r = 1, 2, 3, 4$ (when $\beta > 4$ all exist) we get the explicit raw moments

$$\begin{aligned} \mu'_1 &= E[Y] = w(\alpha\beta) + (1 - w) \frac{\theta}{\beta - 1}, \\ \mu'_2 &= E[Y^2] = w\beta^2\alpha(\alpha + 1) + (1 - w) \frac{\theta(\theta + 1)}{(\beta - 1)(\beta - 2)}, \\ \mu'_3 &= E[Y^3] = w\beta^3\alpha(\alpha + 1)(\alpha + 2) + (1 - w) \frac{\theta(\theta + 1)(\theta + 2)}{(\beta - 1)(\beta - 2)(\beta - 3)}, \\ \mu'_4 &= E[Y^4] = w\beta^4\alpha(\alpha + 1)(\alpha + 2)(\alpha + 3) \\ &\quad + (1 - w) \frac{\theta(\theta + 1)(\theta + 2)(\theta + 3)}{(\beta - 1)(\beta - 2)(\beta - 3)(\beta - 4)}. \end{aligned} \quad (28)$$

Central moments The central moments are obtained from raw moments by

$$\begin{aligned} \mu_1 &= 0, \\ \mu_2 &= E[(Y - \mu_1)^2] = \mu'_2 - (\mu'_1)^2, \\ \mu_3 &= E[(Y - \mu_1)^3] = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3, \\ \mu_4 &= E[(Y - \mu_1)^4] = \mu'_4 - 4\mu'_1\mu'_3 + 6(\mu'_1)^2\mu'_2 - 3(\mu'_1)^4. \end{aligned} \quad (29)$$

Variance We can calculate the variance by raw-moments difference

$$\text{Var}(Y) = \mu'_2 - (\mu'_1)^2,$$

or the law-of-total-variance

$$\text{Var}(Y) = w \text{Var}(Y_1) + (1 - w) \text{Var}(Y_2) + w(1 - w)(\mu_{1,1} - \mu_{2,1})^2 \quad (30)$$

where $\mu_{1,1} = E[Y_1] = \alpha\beta$ and $\mu_{2,1} = E[Y_2] = \frac{\theta}{\beta - 1}$.

Substituting known component variances:

$$\text{Var}(Y_1) = \alpha\beta^2, \quad \text{Var}(Y_2) = \frac{\theta(\theta + \beta - 1)}{(\beta - 2)(\beta - 1)^2} \quad (\beta > 2),$$

yields the explicit form

$$\begin{aligned} Var(Y) &= w\alpha\beta^2 + (1-w)\frac{\theta(\theta+\beta-1)}{(\beta-2)(\beta-1)^2} + w(1-w)\left(\alpha\beta - \frac{\theta}{\beta-1}\right)^2 \\ Var(Y) &= \frac{\alpha\beta}{\alpha\beta+1}(\alpha\beta^2) + \frac{1}{\alpha\beta+1}\frac{\theta(\theta+\beta-1)}{(\beta-2)(\beta-1)^2} + \frac{\alpha\beta}{(\alpha\beta+1)^2}\left(\alpha\beta - \frac{\theta}{\beta-1}\right)^2 \end{aligned} \quad (31)$$

Skewness and kurtosis Using the central moments above, the skewness and kurtosis are

$$\text{Skewness } (\gamma_1) = \frac{\mu_3}{\mu_2^{3/2}}, \quad \text{Kurtosis } (\gamma_2) = \frac{\mu_4}{\mu_2^2}.$$

In expanded raw-moment form, skewness can be written as

$$\gamma_1 = \frac{\mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3}{(\mu'_2 - (\mu'_1)^2)^{3/2}}, \quad (32)$$

and kurtosis as

$$\gamma_2 = \frac{\mu'_4 - 4\mu'_1\mu'_3 + 6(\mu'_1)^2\mu'_2 - 3(\mu'_1)^4}{(\mu'_2 - (\mu'_1)^2)^2}. \quad (33)$$

Moment-existence conditions The mean of the proposed distribution exists for $\beta > 1$, while the variance exists for $\beta > 2$. More generally, the r th raw moment of Y exists only if $\beta > r$, which follows from the corresponding existence condition of the Beta-prime component.

Figure 5 illustrates the three-dimensional surface of the mean of the proposed distribution for a fixed value of $\theta = 4$, highlighting the influence of the remaining parameters on the expected value. Figure 6 presents the corresponding three-dimensional surface of the variance for $\theta = 4$, demonstrating the flexibility of the model in capturing varying dispersion levels.

6 Survival and hazard rate function

The survival function, defined as the probability that a unit survives beyond time y , is given by

$$\begin{aligned} S(y) &= 1 - F(y, \alpha, \beta, \theta) \\ S(y) &= 1 - \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right] \end{aligned} \quad (34)$$

The hazard rate function, also known as the failure rate, describes the instantaneous risk of failure at time y , given survival up to that time, and is defined as

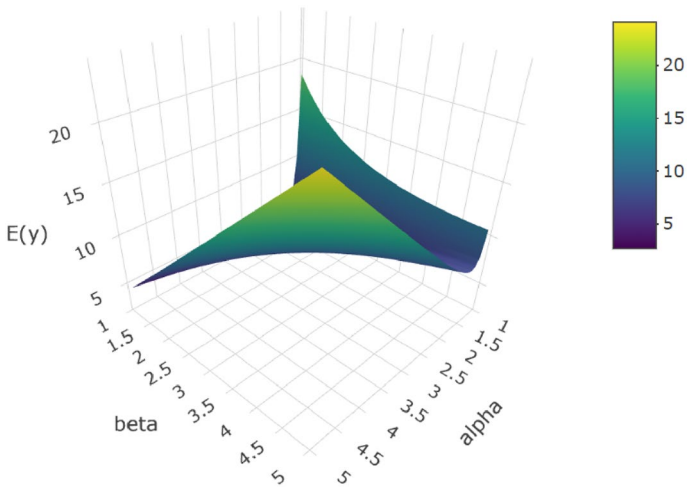


Fig. 5 3D Plot of mean for fixed value of $\theta = 4$

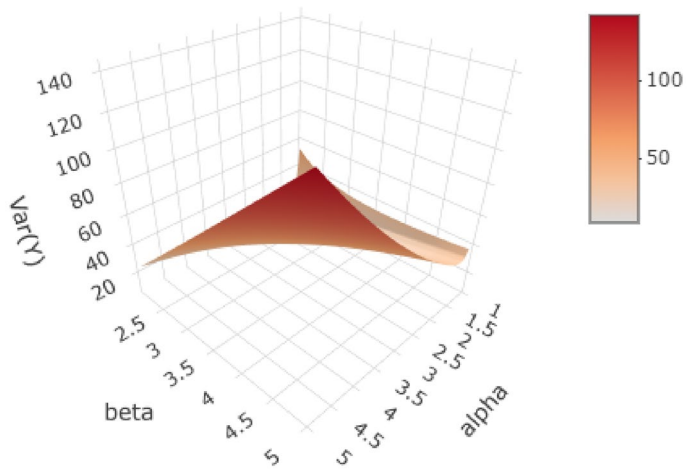


Fig. 6 3D Plot of variance for fixed value of $\theta = 4$

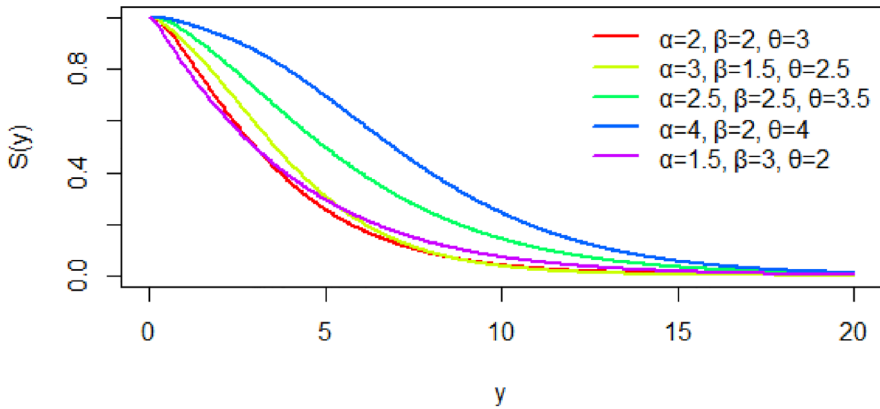


Fig. 7 Plot of the survival function for the mixture distribution

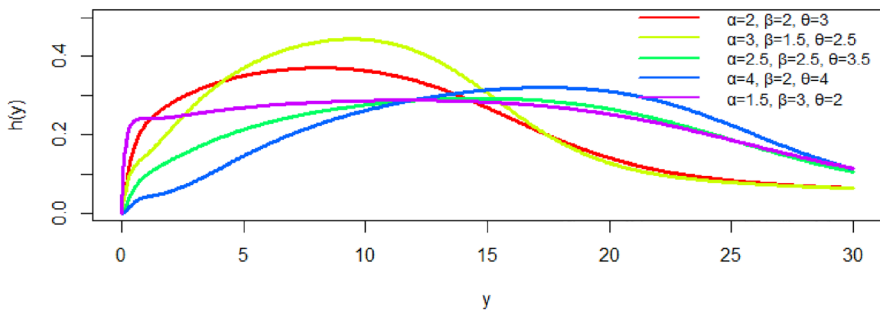


Fig. 8 Plot of the hazard rate function for the mixture distribution

$$h(y) = \frac{f(y)}{S(y)}$$

$$h(y) = \frac{\frac{\alpha\beta}{\alpha\beta+1} \frac{y^{\alpha-1}e^{-y/\beta}}{\beta^\alpha\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} \frac{y^{\theta-1}(1+y)^{-(\theta+\beta)}}{B(\theta, \beta)}}{1 - \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right]} \quad (35)$$

Figures 7 and 8 depict the survival and hazard rate functions of the proposed distribution. The smooth decay of the survival function and the adaptable shape of the hazard rate indicate that the model is capable of capturing diverse lifetime and failure-rate behaviors.

7 Maximum likelihood estimation (MLE)

Let $Y_1, Y_2, \dots, Y_n$ be a random sample of size n drawn from the proposed mixture distribution. The probability density function is given by

$$f(y; \alpha, \theta, \beta) = \frac{\alpha\beta}{\alpha\beta + 1} \frac{y^{\alpha-1} e^{-y/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta + 1} \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)}. \quad (36)$$

where $\alpha > 0$, $\beta > 0$, and $\theta > 0$ are shape and scale parameters. The likelihood function is

$$L(\alpha, \theta, \beta) = \prod_{i=1}^n f(y_i; \alpha, \theta, \beta),$$

and the corresponding log-likelihood function is

$$\begin{aligned} \log L(\alpha, \theta, \beta) &= \sum_{i=1}^n \log f(y_i; \alpha, \theta, \beta). \\ \log L(\alpha, \theta, \beta) &= \sum_{i=1}^n \log \left[\frac{\alpha\beta}{\alpha\beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta + 1} \frac{y_i^{\theta-1} (1+y_i)^{-(\theta+\beta)}}{B(\theta, \beta)} \right] \end{aligned} \quad (37)$$

The partial derivative of the log-likelihood with respect to α is given by

$$\frac{\partial}{\partial \alpha} \log L(\alpha, \theta, \beta) = \sum_{i=1}^n \frac{1}{A_{1,i} + A_{2,i}} \left(\frac{\partial A_{1,i}}{\partial \alpha} + \frac{\partial A_{2,i}}{\partial \alpha} \right)$$

$$A_{1,i} = \frac{\alpha\beta}{\alpha\beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)}$$

$$A_{2,i} = \frac{1}{\alpha\beta + 1} \frac{y_i^{\theta-1} (1+y_i)^{-(\theta+\beta)}}{B(\theta, \beta)}$$

$$\begin{aligned} \frac{\partial A_{1,i}}{\partial \alpha} &= \left[\frac{\beta}{(\alpha\beta + 1)^2} \right] \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \\ &+ \frac{\alpha\beta}{\alpha\beta + 1} \left[y_i^{\alpha-1} \log y_i \frac{e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \right] \\ &+ \frac{\alpha\beta}{\alpha\beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} (-\log \beta) \\ &- \frac{\alpha\beta}{\alpha\beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \psi(\alpha) \end{aligned}$$

$$\frac{\partial A_{2,i}}{\partial \alpha} = - \frac{\beta}{(\alpha\beta + 1)^2} \frac{y_i^{\theta-1} (1+y_i)^{-(\theta+\beta)}}{B(\theta, \beta)}$$

The partial derivative of the log-likelihood with respect to β is:

$$\frac{\partial}{\partial \beta} \log L(\alpha, \theta, \beta) = \sum_{i=1}^n \frac{1}{A_{1,i} + A_{2,i}} \left(\frac{\partial A_{1,i}}{\partial \beta} + \frac{\partial A_{2,i}}{\partial \beta} \right)$$

where

$$A_{1,i} = \frac{\alpha \beta}{\alpha \beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)}$$

$$A_{2,i} = \frac{1}{\alpha \beta + 1} \frac{y_i^{\theta-1} (1 + y_i)^{-(\theta+\beta)}}{B(\theta, \beta)}$$

The derivatives are:

$$\begin{aligned} \frac{\partial A_{1,i}}{\partial \beta} &= \left[\frac{\alpha(\alpha \beta + 1) - \alpha \beta(\alpha)}{(\alpha \beta + 1)^2} \right] \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \\ &\quad + \frac{\alpha \beta}{\alpha \beta + 1} \left[y_i^{\alpha-1} \frac{e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \left(\frac{y_i}{\beta^2} \right) \right] \\ &\quad + \frac{\alpha \beta}{\alpha \beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)} \left(-\frac{\alpha}{\beta} \right) \end{aligned}$$

Similarly, for $A_{2,i}$:

$$\begin{aligned} \frac{\partial A_{2,i}}{\partial \beta} &= -\frac{\alpha}{(\alpha \beta + 1)^2} \frac{y_i^{\theta-1} (1 + y_i)^{-(\theta+\beta)}}{B(\theta, \beta)} \\ &\quad + \frac{1}{\alpha \beta + 1} \frac{y_i^{\theta-1} (1 + y_i)^{-(\theta+\beta)}}{B(\theta, \beta)} (-\log(1 + y_i)) \\ &\quad + \text{additional terms from } B(\theta, \beta). \end{aligned}$$

The derivative of $B(\theta, \beta)$ with respect to β involves digamma functions and leads to lengthy expressions; hence, it is omitted here for brevity.

The partial derivative of the log-likelihood with respect to θ is:

$$\frac{\partial}{\partial \theta} \log L(\alpha, \theta, \beta) = \sum_{i=1}^n \frac{1}{A_{1,i} + A_{2,i}} \left(\frac{\partial A_{1,i}}{\partial \theta} + \frac{\partial A_{2,i}}{\partial \theta} \right)$$

where

$$A_{1,i} = \frac{\alpha \beta}{\alpha \beta + 1} \frac{y_i^{\alpha-1} e^{-y_i/\beta}}{\beta^\alpha \Gamma(\alpha)}$$

$$A_{2,i} = \frac{1}{\alpha \beta + 1} \frac{y_i^{\theta-1} (1 + y_i)^{-(\theta+\beta)}}{B(\theta, \beta)}$$

The derivatives are:

$$\begin{aligned}\frac{\partial A_{1,i}}{\partial \theta} &= 0 \\ \frac{\partial A_{2,i}}{\partial \theta} &= \frac{\partial}{\partial \theta} \left[\frac{1}{\alpha\beta + 1} \frac{y_i^{\theta-1} (1+y_i)^{-(\theta+\beta)}}{B(\theta, \beta)} \right] \\ \frac{\partial A_{2,i}}{\partial \theta} &= \frac{1}{\alpha\beta + 1} \left[\frac{y_i^{\theta-1} (1+y_i)^{-(\theta+\beta)}}{B(\theta, \beta)} (\log y_i - \log(1+y_i) - \psi(\theta) + \psi(\theta + \beta)) \right] \\ \frac{\partial}{\partial \theta} \log L(\alpha, \theta, \beta) &= \sum_{i=1}^n \frac{1}{A_{1,i} + A_{2,i}} \frac{\partial A_{2,i}}{\partial \theta} \\ \frac{\partial A_{2,i}}{\partial \theta} &= A_{2,i} (\log y_i - \log(1+y_i) - \psi(\theta) + \psi(\theta + \beta))\end{aligned}\quad (38)$$

Analytical solutions to the resulting likelihood equations are not available. Therefore, numerical optimization techniques, such as the Newton–Raphson method, are employed to obtain the maximum likelihood estimates.

8 Simulation

An analysis of a GBPD through simulation was carried out to evaluate and compare the efficiency and influence of sample size, mixing proportion and the magnitude of parametric values on these estimates. The simulation study is carried out in R. Random sample sizes $n = 100, 150, 200, 250, 300, 350$ are generated with various combinations of parametric values. As one sample cannot completely describe the behaviors and properties of GBPD, the results have been replicated $N = 1000$ times, and the average of the results is presented in Tables 3, 4, 5, 6 and 7. In this section we evaluate the performance of the estimation methods offered for estimating the parameters of the GBPD model using simulation data. The average absolute biases (BIAS), mean square errors (MSE), and mean relative errors (MRE) are determined by applying the following formulas (Casella and Berger 2002; Lawless 2003).

$$\begin{aligned}\text{Bias}(\hat{\theta}) &= \frac{1}{N} \sum_{i=1}^N (\hat{\theta}_i - \theta) \\ \text{MSE}(\hat{\theta}) &= \frac{1}{N} \sum_{i=1}^N (\hat{\theta}_i - \theta)^2\end{aligned}$$

$$\text{MRE}(\hat{\theta}) = \frac{1}{N} \sum_{i=1}^N \left| \frac{\hat{\theta}_i - \theta}{\theta} \right|$$

where

- $\hat{\theta}_i$: Estimated value of the parameter in the i^{th} simulation
- θ : True parameter value
- N : Number of simulations

We will discuss various conventional parameter estimation methods for the GBPD. These methods involve optimizing an objective function to obtain the parameter estimator.

8.1 Maximum likelihood estimation (EM1)

Maximum likelihood estimator is powerful statistical tool utilize to estimate the parameters of a probability distribution or a statistical model Casella and Berger (2002).

Let $Y_1, Y_2, Y_3, \dots$ be random sample of size n from GBPD then

Likelihood Function can be calculated as

$$L(\theta) = \prod_{i=1}^n f(y_i; \theta) = f(y_1; \theta) f(y_2; \theta) f(y_3; \theta)$$

Log-Likelihood Function

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(y_i; \theta)$$

MLE of θ is the value that maximizes the log-likelihood

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$$

8.2 Anderson-Darling estimation (ADE) (EM2)

It is a statistical method used to assess how well a sample of data fits a specified probability distribution, often used for goodness-of-fit testing. It is based on the ADE test statistic, which gives more weight to the tails of the distribution (Tan 2006).

$$ADE(\theta) = -n - \sum_{i=1}^n \frac{2i-1}{n} [\log F(y_{(i)}; \theta) + \log (1 - F(y_{(n-i+1)}; \theta))]$$

The ADE estimate $\hat{\theta}_{\text{ADE}}$ is obtained by minimizing this criterion:

Table 3 Simulation values of BIAS, MSE, and MRE for $\alpha = 3$, $\beta = 5$, $\theta = 3$, $w = 0.9375$

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
100	BIAS	alpha	-0.1014 ⁽⁴⁾	-2.8314 ⁽¹⁾	-0.0023 ⁽⁶⁾	5.5386 ⁽⁹⁾	-0.7981 ⁽²⁾	0.1038 ⁽⁸⁾	-0.0440 ⁽⁵⁾	-0.3366 ⁽³⁾
		beta	0.0726 ⁽⁷⁾	-2.9125 ⁽²⁾	-0.0521 ⁽⁴⁾	-3.3577 ⁽¹⁾	-0.7625 ⁽³⁾	0.0585 ⁽⁶⁾	0.2133 ⁽⁸⁾	0.2644 ⁽⁹⁾
		theta	1.7159 ⁽⁴⁾	16.1570 ⁽⁹⁾	2.1972 ⁽⁵⁾	9.8723 ⁽⁸⁾	4.6643 ⁽⁷⁾	0.1648 ⁽¹⁾	0.1674 ⁽²⁾	1.6852 ⁽³⁾
	MSE	alpha	0.5966 ⁽⁵⁾	8.4671 ⁽⁸⁾	0.5414 ⁽³⁾	34.7877 ⁽⁹⁾	2.6044 ⁽⁷⁾	0.4986 ⁽¹⁾	0.5396 ⁽²⁾	0.8238 ⁽⁶⁾
		beta	1.7963 ⁽⁵⁾	8.8806 ⁽⁸⁾	1.1197 ⁽¹⁾	11.3998 ⁽⁹⁾	3.3697 ⁽⁷⁾	1.6853 ⁽³⁾	1.7383 ⁽⁴⁾	2.2063 ⁽⁶⁾
		theta	27.3021 ⁽³⁾	274.5824 ⁽⁹⁾	34.5982 ⁽⁵⁾	129.4078 ⁽⁸⁾	79.6396 ⁽⁷⁾	1.5552 ⁽¹⁾	7.3343 ⁽²⁾	32.7453 ⁽⁴⁾
	MRE	alpha	0.1711 ⁽⁴⁾	0.9515 ⁽⁸⁾	0.1588 ⁽¹⁾	1.8661 ⁽⁹⁾	0.3605 ⁽⁷⁾	0.1668 ⁽³⁾	0.1805 ⁽⁵⁾	0.1913 ⁽⁶⁾
		beta	0.1852 ⁽³⁾	0.5886 ⁽⁸⁾	0.1568 ⁽¹⁾	0.6715 ⁽⁹⁾	0.2765 ⁽⁷⁾	0.1888 ⁽⁴⁾	0.2034 ⁽⁵⁾	0.2138 ⁽⁶⁾
		theta	0.7754 ⁽³⁾	5.3951 ⁽⁹⁾	0.9696 ⁽⁵⁾	3.2908 ⁽⁸⁾	1.7488 ⁽⁷⁾	0.2941 ⁽¹⁾	0.4376 ⁽²⁾	0.8736 ⁽⁴⁾
	$\sum$ Ranks		38	62	31	70	54	28	35	47
		alpha	-0.0518 ⁽⁴⁾	-2.0964 ⁽¹⁾	0.0545 ⁽⁵⁾	4.8886 ⁽⁹⁾	-0.7265 ⁽²⁾	0.1475 ⁽⁸⁾	0.0952 ⁽⁷⁾	-0.2036 ⁽³⁾
		beta	-0.0937 ⁽⁷⁾	-2.0044 ⁽²⁾	-0.1271 ⁽⁴⁾	-3.2012 ⁽¹⁾	-0.6960 ⁽³⁾	-0.1039 ⁽⁶⁾	0.0308 ⁽⁹⁾	-0.0023 ⁽⁸⁾
150	BIAS	alpha	0.0726 ⁽⁷⁾	16.1570 ⁽⁹⁾	2.1972 ⁽⁵⁾	9.8723 ⁽⁸⁾	4.6643 ⁽⁷⁾	0.1648 ⁽¹⁾	0.1674 ⁽²⁾	1.6852 ⁽³⁾
		beta	1.7159 ⁽⁴⁾	8.4671 ⁽⁸⁾	0.5414 ⁽³⁾	34.7877 ⁽⁹⁾	2.6044 ⁽⁷⁾	0.4986 ⁽¹⁾	0.5396 ⁽²⁾	0.8238 ⁽⁶⁾
		theta	27.3021 ⁽³⁾	274.5824 ⁽⁹⁾	34.5982 ⁽⁵⁾	129.4078 ⁽⁸⁾	79.6396 ⁽⁷⁾	1.5552 ⁽¹⁾	7.3343 ⁽²⁾	32.7453 ⁽⁴⁾
	MSE	alpha	0.1711 ⁽⁴⁾	0.9515 ⁽⁸⁾	0.1588 ⁽¹⁾	1.8661 ⁽⁹⁾	0.3605 ⁽⁷⁾	0.1668 ⁽³⁾	0.1805 ⁽⁵⁾	0.1913 ⁽⁶⁾
		beta	0.1852 ⁽³⁾	0.5886 ⁽⁸⁾	0.1568 ⁽¹⁾	0.6715 ⁽⁹⁾	0.2765 ⁽⁷⁾	0.1888 ⁽⁴⁾	0.2034 ⁽⁵⁾	0.2138 ⁽⁶⁾
		theta	0.7754 ⁽³⁾	5.3951 ⁽⁹⁾	0.9696 ⁽⁵⁾	3.2908 ⁽⁸⁾	1.7488 ⁽⁷⁾	0.2941 ⁽¹⁾	0.4376 ⁽²⁾	0.8736 ⁽⁴⁾
	MRE	alpha	0.1711 ⁽⁴⁾	0.9515 ⁽⁸⁾	0.1588 ⁽¹⁾	1.8661 ⁽⁹⁾	0.3605 ⁽⁷⁾	0.1668 ⁽³⁾	0.1805 ⁽⁵⁾	0.1913 ⁽⁶⁾
		beta	0.1852 ⁽³⁾	0.5886 ⁽⁸⁾	0.1568 ⁽¹⁾	0.6715 ⁽⁹⁾	0.2765 ⁽⁷⁾	0.1888 ⁽⁴⁾	0.2034 ⁽⁵⁾	0.2138 ⁽⁶⁾
		theta	0.7754 ⁽³⁾	5.3951 ⁽⁹⁾	0.9696 ⁽⁵⁾	3.2908 ⁽⁸⁾	1.7488 ⁽⁷⁾	0.2941 ⁽¹⁾	0.4376 ⁽²⁾	0.8736 ⁽⁴⁾
	$\sum$ Ranks		38	62	31	70	54	28	35	47
		alpha	-0.0518 ⁽⁴⁾	-2.0964 ⁽¹⁾	0.0545 ⁽⁵⁾	4.8886 ⁽⁹⁾	-0.7265 ⁽²⁾	0.1475 ⁽⁸⁾	0.0952 ⁽⁷⁾	-0.2036 ⁽³⁾
		beta	-0.0937 ⁽⁷⁾	-2.0044 ⁽²⁾	-0.1271 ⁽⁴⁾	-3.2012 ⁽¹⁾	-0.6960 ⁽³⁾	-0.1039 ⁽⁶⁾	0.0308 ⁽⁹⁾	-0.0023 ⁽⁸⁾
200	BIAS	alpha	0.0726 ⁽⁷⁾	16.1570 ⁽⁹⁾	2.1972 ⁽⁵⁾	9.8723 ⁽⁸⁾	4.6643 ⁽⁷⁾	0.1648 ⁽¹⁾	0.1674 ⁽²⁾	1.6852 ⁽³⁾
		beta	1.7159 ⁽⁴⁾	8.4671 ⁽⁸⁾	0.5414 ⁽³⁾	34.7877 ⁽⁹⁾	2.6044 ⁽⁷⁾	0.4986 ⁽¹⁾	0.5396 ⁽²⁾	0.8238 ⁽⁶⁾
		theta	27.3021 ⁽³⁾	274.5824 ⁽⁹⁾	34.5982 ⁽⁵⁾	129.4078 ⁽⁸⁾	79.6396 ⁽⁷⁾	1.5552 ⁽¹⁾	7.3343 ⁽²⁾	32.7453 ⁽⁴⁾
	MSE	alpha	0.1711 ⁽⁴⁾	0.9515 ⁽⁸⁾	0.1588 ⁽¹⁾	1.8661 ⁽⁹⁾	0.3605 ⁽⁷⁾	0.1668 ⁽³⁾	0.1805 ⁽⁵⁾	0.1913 ⁽⁶⁾
		beta	0.1852 ⁽³⁾	0.5886 ⁽⁸⁾	0.1568 ⁽¹⁾	0.6715 ⁽⁹⁾	0.2765 ⁽⁷⁾	0.1888 ⁽⁴⁾	0.2034 ⁽⁵⁾	0.2138 ⁽⁶⁾
		theta	0.7754 ⁽³⁾	5.3951 ⁽⁹⁾	0.9696 ⁽⁵⁾	3.2908 ⁽⁸⁾	1.7488 ⁽⁷⁾	0.2941 ⁽¹⁾	0.4376 ⁽²⁾	0.8736 ⁽⁴⁾
	MRE	alpha	0.1711 ⁽⁴⁾	0.9515 ⁽⁸⁾	0.1588 ⁽¹⁾	1.8661 ⁽⁹⁾	0.3605 ⁽⁷⁾	0.1668 ⁽³⁾	0.1805 ⁽⁵⁾	0.1913 ⁽⁶⁾
		beta	0.1852 ⁽³⁾	0.5886 ⁽⁸⁾	0.1568 ⁽¹⁾	0.6715 ⁽⁹⁾	0.2765 ⁽⁷⁾	0.1888 ⁽⁴⁾	0.2034 ⁽⁵⁾	0.2138 ⁽⁶⁾
		theta	0.7754 ⁽³⁾	5.3951 ⁽⁹⁾	0.9696 ⁽⁵⁾	3.2908 ⁽⁸⁾	1.7488 ⁽⁷⁾	0.2941 ⁽¹⁾	0.4376 ⁽²⁾	0.8736 ⁽⁴⁾
	$\sum$ Ranks		38	62	31	70	54	28	35	47
		alpha	-0.0518 ⁽⁴⁾	-2.0964 ⁽¹⁾	0.0545 ⁽⁵⁾	4.8886 ⁽⁹⁾	-0.7265 ⁽²⁾	0.1475 ⁽⁸⁾	0.0952 ⁽⁷⁾	-0.2036 ⁽³⁾
		beta	-0.0937 ⁽⁷⁾	-2.0044 ⁽²⁾	-0.1271 ⁽⁴⁾	-3.2012 ⁽¹⁾	-0.6960 ⁽³⁾	-0.1039 ⁽⁶⁾	0.0308 ⁽⁹⁾	-0.0023 ⁽⁸⁾

Table 3 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
250	$\sum$ Ranks	beta	0.1016 ⁽¹⁾	0.3395 ⁽⁸⁾	0.1196 ⁽³⁾	0.5992 ⁽⁹⁾	0.1806 ⁽⁷⁾	0.1309 ⁽⁶⁾	0.1269 ⁽⁵⁾	0.1019 ⁽²⁾
		theta	0.2110 ⁽³⁾	2.6324 ⁽⁸⁾	0.5584 ⁽⁵⁾	3.4141 ⁽⁹⁾	1.1093 ⁽⁷⁾	0.1704 ⁽¹⁾	0.2108 ⁽²⁾	0.2339 ⁽⁴⁾
	BIAS	alpha	29	59	35	73	54	40	38	33
		beta	0.0370 ⁽⁷⁾	-0.7340 ⁽¹⁾	-0.0044 ⁽⁴⁾	3.2404 ⁽⁹⁾	-0.4797 ⁽²⁾	0.0737 ⁽⁸⁾	0.0034 ⁽⁶⁾	-0.0308 ⁽³⁾
	MSE	theta	-0.0172 ⁽⁵⁾	-0.6292 ⁽²⁾	-0.0084 ⁽⁶⁾	-2.7138 ⁽¹⁾	-0.3856 ⁽³⁾	-0.0192 ⁽⁴⁾	0.0847 ⁽⁸⁾	0.1150 ⁽⁹⁾
		alpha	0.1230 ⁽⁴⁾	4.1256 ⁽⁸⁾	1.5088 ⁽⁵⁾	13.1392 ⁽⁹⁾	3.1182 ⁽⁷⁾	0.0507 ⁽³⁾	-0.0818 ⁽²⁾	-0.1005 ⁽¹⁾
	MRE	beta	0.0987 ⁽²⁾	2.1129 ⁽⁸⁾	0.2360 ⁽⁵⁾	12.9127 ⁽⁹⁾	1.4341 ⁽⁷⁾	0.2169 ⁽⁴⁾	0.1531 ⁽³⁾	0.0933 ⁽¹⁾
		theta	0.2575 ⁽¹⁾	2.4242 ⁽⁸⁾	0.4940 ⁽⁴⁾	7.5890 ⁽⁹⁾	1.8125 ⁽⁷⁾	0.6243 ⁽⁶⁾	0.4792 ⁽³⁾	0.2848 ⁽²⁾
	$\sum$ Ranks	alpha	0.4383 ⁽²⁾	67.0700 ⁽⁸⁾	20.9697 ⁽⁵⁾	216.0428 ⁽⁹⁾	48.7226 ⁽⁷⁾	0.5870 ⁽³⁾	0.7274 ⁽⁴⁾	0.3862 ⁽¹⁾
		beta	0.0811 ⁽²⁾	0.2916 ⁽⁸⁾	0.1104 ⁽⁴⁾	1.0801 ⁽⁹⁾	0.2274 ⁽⁷⁾	0.1244 ⁽⁶⁾	0.1038 ⁽³⁾	0.0802 ⁽¹⁾
	BIAS	theta	0.0826 ⁽¹⁾	0.2109 ⁽⁸⁾	0.1069 ⁽³⁾	0.5428 ⁽⁹⁾	0.1812 ⁽⁷⁾	0.1284 ⁽⁶⁾	0.1074 ⁽⁵⁾	0.0851 ⁽²⁾
		alpha	0.1662 ⁽¹⁾	1.4636 ⁽⁸⁾	0.6602 ⁽⁵⁾	4.3984 ⁽⁹⁾	1.2079 ⁽⁷⁾	0.1911 ⁽³⁾	0.2229 ⁽⁴⁾	0.1664 ⁽²⁾
300	$\sum$ Ranks	beta	25	59	41	73	54	43	38	22
		theta	0.0431 ⁽⁵⁾	-0.5313 ⁽¹⁾	0.0588 ⁽⁶⁾	2.7004 ⁽⁹⁾	-0.4306 ⁽²⁾	0.0704 ⁽⁸⁾	0.0064 ⁽⁴⁾	-0.0154 ⁽³⁾
	BIAS	alpha	-0.0214 ⁽⁶⁾	-0.5530 ⁽²⁾	-0.0308 ⁽⁴⁾	-2.5306 ⁽¹⁾	-0.3823 ⁽³⁾	-0.0205 ⁽⁷⁾	0.0828 ⁽⁸⁾	0.0931 ⁽⁹⁾
		beta	0.0192 ⁽⁴⁾	3.1075 ⁽⁸⁾	0.5138 ⁽⁵⁾	14.4594 ⁽⁹⁾	2.8710 ⁽⁷⁾	-0.0562 ⁽³⁾	-0.1134 ⁽²⁾	-0.1567 ⁽¹⁾
	MSE	theta	0.0655 ⁽²⁾	1.6595 ⁽⁸⁾	0.1053 ⁽³⁾	8.4934 ⁽⁹⁾	1.3127 ⁽⁷⁾	0.1567 ⁽⁶⁾	0.1176 ⁽⁵⁾	0.0621 ⁽¹⁾
		alpha	0.1842 ⁽¹⁾	1.8726 ⁽⁸⁾	0.2897 ⁽³⁾	6.5534 ⁽⁹⁾	1.6019 ⁽⁷⁾	0.4576 ⁽⁶⁾	0.3641 ⁽⁵⁾	0.2035 ⁽²⁾
	MRE	beta	0.3460 ⁽²⁾	52.4862 ⁽⁸⁾	8.0869 ⁽⁵⁾	244.0630 ⁽⁹⁾	44.5015 ⁽⁷⁾	0.3941 ⁽³⁾	0.4799 ⁽⁴⁾	0.3254 ⁽¹⁾
		theta	0.0667 ⁽²⁾	0.2341 ⁽⁸⁾	0.0860 ⁽³⁾	0.9001 ⁽⁹⁾	0.2029 ⁽⁷⁾	0.1003 ⁽⁶⁾	0.0904 ⁽⁵⁾	0.0648 ⁽¹⁾
	$\sum$ Ranks	alpha	0.0670 ⁽¹⁾	0.1679 ⁽⁸⁾	0.0856 ⁽³⁾	0.5061 ⁽⁹⁾	0.1579 ⁽⁷⁾	0.1006 ⁽⁶⁾	0.0947 ⁽⁵⁾	0.0703 ⁽²⁾
		beta	0.1470 ⁽¹⁾	1.1646 ⁽⁸⁾	0.3746 ⁽⁵⁾	4.8710 ⁽⁹⁾	1.1483 ⁽⁷⁾	0.1575 ⁽³⁾	0.1853 ⁽⁴⁾	0.1518 ⁽²⁾
	BIAS	theta	24	59	37	73	59	46	42	22
		alpha	-0.0106 ⁽⁷⁾	-0.2254 ⁽²⁾	-0.0331 ⁽⁵⁾	2.3426 ⁽⁹⁾	-0.3369 ⁽¹⁾	-0.0054 ⁽⁸⁾	-0.0339 ⁽⁴⁾	-0.0598 ⁽³⁾
350	$\sum$ Ranks	beta	0.0582 ⁽⁴⁾	-0.0778 ⁽³⁾	0.1076 ⁽⁶⁾	-2.4366 ⁽¹⁾	-0.1370 ⁽²⁾	0.0784 ⁽⁵⁾	0.1390 ⁽⁸⁾	0.1568 ⁽⁹⁾
		theta	0.0899 ⁽⁴⁾	1.0924 ⁽⁷⁾	0.3745 ⁽⁵⁾	15.9732 ⁽⁹⁾	1.9687 ⁽⁸⁾	0.0259 ⁽³⁾	-0.0125 ⁽²⁾	-0.0635 ⁽¹⁾
	MSE	alpha	0.0726 ⁽¹⁾	0.6154 ⁽⁷⁾	0.1034 ⁽³⁾	5.7605 ⁽⁹⁾	0.8853 ⁽⁸⁾	0.1358 ⁽⁶⁾	0.1149 ⁽⁵⁾	0.0739 ⁽²⁾
		beta								

Table 3 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
MRE		beta	0.2003 ⁽¹⁾	0.8191 ⁽⁷⁾	0.3145 ⁽³⁾	5.9873 ⁽⁹⁾	1.1930 ⁽⁸⁾	0.4384 ⁽⁶⁾	0.3958 ⁽⁵⁾	0.2297 ⁽²⁾
		theta	0.2296 ⁽²⁾	17.6354 ⁽⁷⁾	2.6214 ⁽⁵⁾	271.7717 ⁽⁹⁾	30.5829 ⁽⁸⁾	0.2730 ⁽³⁾	0.3285 ⁽⁴⁾	0.2174 ⁽¹⁾
		alpha	0.0713 ⁽¹⁾	0.1358 ⁽⁷⁾	0.0842 ⁽³⁾	0.7809 ⁽⁹⁾	0.1624 ⁽⁸⁾	0.0952 ⁽⁶⁾	0.0888 ⁽⁵⁾	0.0736 ⁽²⁾
		beta	0.0681 ⁽¹⁾	0.1133 ⁽⁷⁾	0.0888 ⁽³⁾	0.4873 ⁽⁹⁾	0.1378 ⁽⁸⁾	0.1016 ⁽⁶⁾	0.0964 ⁽⁵⁾	0.0741 ⁽²⁾
$\sum$ Ranks		theta	0.1195 ⁽²⁾	0.4765 ⁽⁷⁾	0.2751 ⁽⁵⁾	5.3486 ⁽⁹⁾	0.8846 ⁽⁸⁾	0.1306 ⁽³⁾	0.1496 ⁽⁴⁾	0.1188 ⁽¹⁾
			23	54	38	73	59	46	42	23

Table 4 Simulation values of BIAS, MSE, and MRE for $\alpha = 3$, $\beta = 2$, $\theta = 3$, $w = 0.86$

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
100	BIAS	alpha	0.0838 ⁽¹⁾	0.1179 ⁽²⁾	0.1437 ⁽³⁾	1.2780 ⁽⁹⁾	0.1809 ⁽⁷⁾	0.1866 ⁽⁸⁾	0.1698 ⁽⁶⁾	0.1633 ⁽⁵⁾
		beta	-0.0080 ⁽⁹⁾	-0.0103 ⁽⁸⁾	-0.0240 ⁽⁶⁾	-0.6546 ⁽¹⁾	-0.0372 ⁽³⁾	-0.0273 ⁽⁵⁾	-0.0448 ⁽²⁾	-0.0309 ⁽⁴⁾
	MSE	theta	0.3254 ⁽⁵⁾	0.5219 ⁽⁶⁾	0.6872 ⁽⁸⁾	-1.5129 ⁽¹⁾	0.5854 ⁽⁷⁾	0.1518 ⁽⁴⁾	0.0158 ⁽²⁾	0.0489 ⁽³⁾
		alpha	0.4118 ⁽¹⁾	0.5414 ⁽²⁾	0.5666 ⁽⁵⁾	1.9089 ⁽⁹⁾	0.6597 ⁽⁷⁾	0.7214 ⁽⁸⁾	0.5423 ⁽³⁾	0.5651 ⁽⁴⁾
	MRE	beta	0.1611 ⁽¹⁾	0.2284 ⁽³⁾	0.2680 ⁽⁵⁾	0.4532 ⁽⁹⁾	0.2756 ⁽⁷⁾	0.2770 ⁽⁸⁾	0.2295 ⁽⁴⁾	0.2187 ⁽²⁾
		theta	1.3599 ⁽¹⁾	4.8961 ⁽⁶⁾	5.7960 ⁽⁸⁾	3.8296 ⁽⁵⁾	5.5151 ⁽⁷⁾	1.5579 ⁽²⁾	2.3837 ⁽⁴⁾	2.1500 ⁽³⁾
	Ranks	alpha	0.1636 ⁽¹⁾	0.1951 ⁽⁴⁾	0.2056 ⁽⁵⁾	0.4260 ⁽⁹⁾	0.2173 ⁽⁸⁾	0.2094 ⁽⁷⁾	0.1931 ⁽³⁾	0.1884 ⁽²⁾
		beta	0.1614 ⁽¹⁾	0.1998 ⁽³⁾	0.2174 ⁽⁶⁾	0.3273 ⁽⁹⁾	0.2210 ⁽⁸⁾	0.2116 ⁽⁵⁾	0.2031 ⁽⁴⁾	0.1905 ⁽²⁾
	BIAS	theta	0.2865 ⁽¹⁾	0.4111 ⁽⁵⁾	0.5211 ⁽⁷⁾	0.5811 ⁽⁹⁾	0.5013 ⁽⁶⁾	0.3048 ⁽³⁾	0.3310 ⁽⁴⁾	0.2993 ⁽²⁾
			21	39	53	61	60	50	32	27
150	BIAS	alpha	-0.0352 ⁽¹⁾	-0.0164 ⁽²⁾	-0.0120 ⁽³⁾	1.1677 ⁽⁹⁾	-0.0007 ⁽⁵⁾	0.0283 ⁽⁶⁾	0.0840 ⁽⁸⁾	0.0543 ⁽⁷⁾
		beta	0.0608 ⁽⁴⁾	0.0629 ⁽⁶⁾	0.0691 ⁽⁸⁾	-0.5777 ⁽¹⁾	0.0664 ⁽⁷⁾	0.0618 ⁽⁵⁾	-0.0070 ⁽²⁾	0.0049 ⁽³⁾
	MSE	theta	0.3253 ⁽⁵⁾	0.4850 ⁽⁶⁾	0.7371 ⁽⁷⁾	-1.1692 ⁽¹⁾	0.7658 ⁽⁹⁾	0.2454 ⁽⁴⁾	0.0645 ⁽²⁾	0.0753 ⁽³⁾
		alpha	0.1725 ⁽¹⁾	0.2606 ⁽³⁾	0.3322 ⁽⁵⁾	1.8874 ⁽⁹⁾	0.3624 ⁽⁸⁾	0.3421 ⁽⁷⁾	0.3133 ⁽⁴⁾	0.2201 ⁽²⁾
	MRE	beta	0.0829 ⁽¹⁾	0.1300 ⁽³⁾	0.1664 ⁽⁵⁾	0.3795 ⁽⁹⁾	0.1785 ⁽⁷⁾	0.2039 ⁽⁸⁾	0.1415 ⁽⁴⁾	0.1007 ⁽²⁾
		theta	1.1892 ⁽¹⁾	3.6275 ⁽⁵⁾	5.8901 ⁽⁶⁾	6.2177 ⁽⁹⁾	6.1979 ⁽⁸⁾	1.3316 ⁽²⁾	1.8670 ⁽⁴⁾	1.6557 ⁽³⁾
	Ranks	alpha	0.1128 ⁽¹⁾	0.1361 ⁽³⁾	0.1513 ⁽⁴⁾	0.3942 ⁽⁹⁾	0.1593 ⁽⁸⁾	0.1562 ⁽⁷⁾	0.1519 ⁽⁶⁾	0.1245 ⁽²⁾
		beta	0.1159 ⁽¹⁾	0.1423 ⁽³⁾	0.1609 ⁽⁵⁾	0.2914 ⁽⁹⁾	0.1699 ⁽⁷⁾	0.1758 ⁽⁸⁾	0.1517 ⁽⁴⁾	0.1285 ⁽²⁾
	BIAS	theta	0.2357 ⁽²⁾	0.3445 ⁽⁵⁾	0.4772 ⁽⁶⁾	0.6258 ⁽⁹⁾	0.4817 ⁽⁸⁾	0.2622 ⁽³⁾	0.2651 ⁽⁴⁾	0.2328 ⁽¹⁾
			17	36	49	65	67	50	38	25
200	BIAS	alpha	-0.0095 ⁽¹⁾	0.0245 ⁽⁵⁾	-0.0047 ⁽²⁾	1.1148 ⁽⁹⁾	0.0151 ⁽⁴⁾	0.0357 ⁽⁶⁾	0.0658 ⁽⁸⁾	0.0654 ⁽⁷⁾
		beta	0.0390 ⁽⁵⁾	0.0254 ⁽⁴⁾	0.0588 ⁽⁸⁾	-0.5314 ⁽¹⁾	0.0521 ⁽⁷⁾	0.0432 ⁽⁶⁾	0.0120 ⁽³⁾	-0.0106 ⁽²⁾
	MSE	theta	0.3463 ⁽⁵⁾	0.2179 ⁽⁴⁾	0.7839 ⁽⁸⁾	-1.4718 ⁽¹⁾	0.5600 ⁽⁷⁾	0.3485 ⁽⁶⁾	0.0598 ⁽²⁾	0.1235 ⁽³⁾
		alpha	0.1381 ⁽¹⁾	0.1799 ⁽³⁾	0.2712 ⁽⁵⁾	1.6875 ⁽⁹⁾	0.2981 ⁽⁸⁾	0.2951 ⁽⁷⁾	0.2288 ⁽⁴⁾	0.1694 ⁽²⁾
	Ranks	beta	0.0758 ⁽¹⁾	0.0888 ⁽³⁾	0.1520 ⁽⁶⁾	0.3375 ⁽⁹⁾	0.1493 ⁽⁵⁾	0.1580 ⁽⁸⁾	0.1170 ⁽⁴⁾	0.0800 ⁽²⁾
		theta	2.5138 ⁽³⁾	1.2897 ⁽¹⁾	6.6972 ⁽⁸⁾	5.3197 ⁽⁷⁾	3.6686 ⁽⁶⁾	2.9141 ⁽⁵⁾	1.8429 ⁽²⁾	2.8469 ⁽⁴⁾
	BIAS	alpha	0.0966 ⁽¹⁾	0.1120 ⁽³⁾	0.1347 ⁽⁶⁾	0.3739 ⁽⁹⁾	0.1388 ⁽⁸⁾	0.1324 ⁽⁵⁾	0.1275 ⁽⁴⁾	0.1080 ⁽²⁾
		beta								
	MSE	theta								
		alpha								

Table 4 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
250	$\sum$ Ranks	beta	0.1048 ⁽¹⁾	0.1211 ⁽³⁾	0.1507 ⁽⁶⁾	0.2669 ⁽⁹⁾	0.1507 ⁽⁶⁾	0.1450 ⁽⁵⁾	0.1324 ⁽⁴⁾	0.1138 ⁽²⁾
		theta	0.2522 ⁽¹⁾	0.2552 ⁽²⁾	0.4821 ⁽⁷⁾	0.6181 ⁽⁹⁾	0.4001 ⁽⁶⁾	0.2991 ⁽⁵⁾	0.2649 ⁽⁴⁾	0.2596 ⁽³⁾
	BIAS	alpha	19	28	56	63	57	53	35	27
		beta	-0.0455 ⁽¹⁾	-0.0242 ⁽⁴⁾	-0.0245 ⁽²⁾	1.1008 ⁽⁹⁾	-0.0232 ⁽⁵⁾	-0.0208 ⁽⁶⁾	0.0195 ⁽⁸⁾	0.0138 ⁽⁷⁾
	MSE	beta	0.0456 ⁽⁸⁾	0.0362 ⁽⁴⁾	0.0426 ⁽⁶⁾	-0.5204 ⁽¹⁾	0.0421 ⁽⁵⁾	0.0505 ⁽³⁾	0.0126 ⁽³⁾	0.0088 ⁽²⁾
		theta	0.2218 ⁽⁶⁾	0.2128 ⁽⁵⁾	0.3422 ⁽⁷⁾	-1.7032 ⁽¹⁾	0.3818 ⁽⁹⁾	0.1010 ⁽⁴⁾	0.0815 ⁽³⁾	0.0189 ⁽²⁾
	MRE	alpha	0.1071 ⁽¹⁾	0.1266 ⁽³⁾	0.1621 ⁽⁴⁾	1.7316 ⁽⁹⁾	0.1746 ⁽⁶⁾	0.1748 ⁽⁷⁾	0.1861 ⁽⁸⁾	0.1249 ⁽²⁾
		beta	0.0639 ⁽¹⁾	0.0806 ⁽³⁾	0.1036 ⁽⁵⁾	0.3392 ⁽⁹⁾	0.1055 ⁽⁷⁾	0.1196 ⁽⁸⁾	0.0885 ⁽⁴⁾	0.0740 ⁽²⁾
		theta	1.0962 ⁽³⁾	2.1889 ⁽⁵⁾	2.9033 ⁽⁶⁾	3.8857 ⁽⁹⁾	3.3516 ⁽⁸⁾	0.5831 ⁽⁴⁾	0.8468 ⁽²⁾	1.2407 ⁽⁴⁾
		alpha	0.0860 ⁽¹⁾	0.0916 ⁽²⁾	0.1058 ⁽⁴⁾	0.3700 ⁽⁹⁾	0.1108 ⁽⁷⁾	0.1066 ⁽⁶⁾	0.1205 ⁽⁸⁾	0.0864 ⁽⁴⁾
		beta	0.0943 ⁽¹⁾	0.1077 ⁽³⁾	0.1223 ⁽⁴⁾	0.2626 ⁽⁹⁾	0.1247 ⁽⁶⁾	0.1274 ⁽⁸⁾	0.1250 ⁽⁷⁾	0.1034 ⁽²⁾
		theta	0.1874 ⁽²⁾	0.2386 ⁽⁵⁾	0.3307 ⁽⁶⁾	0.5931 ⁽⁹⁾	0.3371 ⁽⁸⁾	0.1759 ⁽³⁾	0.2588 ⁽⁴⁾	0.1537 ⁽¹⁾
300	$\sum$ Ranks	alpha	24	34	44	65	61	50	47	27
		beta	0.0420 ⁽⁴⁾	0.0509 ⁽⁶⁾	0.0393 ⁽¹⁾	0.8841 ⁽⁹⁾	0.0395 ⁽³⁾	0.0488 ⁽⁵⁾	0.0714 ⁽⁷⁾	0.1086 ⁽⁸⁾
	BIAS	beta	-0.0147 ⁽⁴⁾	-0.0152 ⁽³⁾	-0.0059 ⁽⁸⁾	-0.4249 ⁽¹⁾	-0.0081 ⁽⁶⁾	-0.0097 ⁽⁵⁾	-0.0063 ⁽⁷⁾	-0.0588 ⁽²⁾
		theta	0.1044 ⁽⁵⁾	0.0900 ⁽³⁾	0.2717 ⁽⁷⁾	-1.0325 ⁽¹⁾	0.3495 ⁽⁹⁾	0.0996 ⁽⁴⁾	0.1729 ⁽⁶⁾	-0.0781 ⁽²⁾
	MSE	alpha	0.0784 ⁽¹⁾	0.1006 ⁽²⁾	0.1215 ⁽⁵⁾	1.4151 ⁽⁹⁾	0.1301 ⁽⁷⁾	0.1177 ⁽⁴⁾	0.2086 ⁽⁸⁾	0.1008 ⁽³⁾
		beta	0.0401 ⁽¹⁾	0.0492 ⁽³⁾	0.0588 ⁽⁴⁾	0.2800 ⁽⁹⁾	0.0607 ⁽⁶⁾	0.0620 ⁽⁷⁾	0.1045 ⁽⁸⁾	0.0478 ⁽²⁾
	MRE	theta	0.3522 ⁽²⁾	0.4391 ⁽³⁾	2.0303 ⁽⁶⁾	5.9360 ⁽⁹⁾	2.6785 ⁽⁸⁾	0.4723 ⁽⁴⁾	1.6378 ⁽⁵⁾	0.3474 ⁽¹⁾
		alpha	0.0748 ⁽¹⁾	0.0826 ⁽²⁾	0.0927 ⁽⁵⁾	0.3083 ⁽⁹⁾	0.0948 ⁽⁷⁾	0.0833 ⁽³⁾	0.1205 ⁽⁸⁾	0.0864 ⁽⁴⁾
		beta	0.0857 ⁽¹⁾	0.0890 ⁽²⁾	0.0974 ⁽⁴⁾	0.2257 ⁽⁹⁾	0.0991 ⁽⁶⁾	0.0999 ⁽⁷⁾	0.1335 ⁽⁸⁾	0.0941 ⁽³⁾
		theta	0.1498 ⁽¹⁾	0.1742 ⁽⁴⁾	0.2777 ⁽⁶⁾	0.5836 ⁽⁹⁾	0.2998 ⁽⁸⁾	0.1633 ⁽³⁾	0.2588 ⁽⁵⁾	0.1537 ⁽²⁾
	$\sum$ Ranks	alpha	20	28	46	65	60	42	62	27
		beta	-0.0242 ⁽⁴⁾	-0.0219 ⁽⁵⁾	-0.0504 ⁽²⁾	0.7035 ⁽⁹⁾	-0.0627 ⁽¹⁾	0.0024 ⁽⁶⁾	0.0593 ⁽⁸⁾	0.0371 ⁽⁷⁾
350	BIAS	beta	0.0373 ⁽⁵⁾	0.0448 ⁽⁶⁾	0.0738 ⁽⁷⁾	-0.3345 ⁽¹⁾	0.0846 ⁽⁹⁾	0.0304 ⁽⁴⁾	-0.0067 ⁽²⁾	-0.0032 ⁽³⁾
		theta	0.1457 ⁽⁴⁾	0.2500 ⁽⁶⁾	0.5782 ⁽⁷⁾	-0.6479 ⁽¹⁾	0.6870 ⁽⁹⁾	0.1575 ⁽⁵⁾	0.0639 ⁽³⁾	-0.0105 ⁽²⁾
	MSE	alpha	0.0733 ⁽¹⁾	0.1163 ⁽³⁾	0.1784 ⁽⁶⁾	1.2352 ⁽⁹⁾	0.1844 ⁽⁸⁾	0.1265 ⁽⁴⁾	0.1568 ⁽⁵⁾	0.0906 ⁽²⁾
		beta								

Table 4 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
MRE		beta	0.0429 ⁽¹⁾	0.0661 ⁽³⁾	0.0988 ⁽⁶⁾	0.2212 ⁽⁹⁾	0.1074 ⁽⁸⁾	0.0743 ⁽⁴⁾	0.0804 ⁽⁵⁾	0.0503 ⁽²⁾
		theta	0.2804 ⁽²⁾	1.4408 ⁽⁵⁾	3.3735 ⁽⁶⁾	9.2132 ⁽⁹⁾	4.3337 ⁽⁸⁾	0.7161 ⁽⁴⁾	0.6465 ⁽³⁾	0.2514 ⁽¹⁾
		alpha	0.0700 ⁽¹⁾	0.0872 ⁽³⁾	0.1106 ⁽⁶⁾	0.2666 ⁽⁹⁾	0.1130 ⁽⁸⁾	0.0920 ⁽⁴⁾	0.1031 ⁽⁵⁾	0.0787 ⁽²⁾
		beta	0.0781 ⁽¹⁾	0.0994 ⁽³⁾	0.1267 ⁽⁶⁾	0.1925 ⁽⁹⁾	0.1308 ⁽⁸⁾	0.1099 ⁽⁵⁾	0.1095 ⁽⁴⁾	0.0854 ⁽²⁾
		theta	0.1379 ⁽²⁾	0.1934 ⁽⁵⁾	0.3512 ⁽⁶⁾	0.7018 ⁽⁹⁾	0.3763 ⁽⁸⁾	0.1676 ⁽³⁾	0.1882 ⁽⁴⁾	0.1321 ⁽¹⁾
$\sum$ Ranks			21	39	52	65	67	39	39	22

Table 5 Simulation values of BIAS, MSE, and MRE for $\alpha = 2, \beta = 3, \theta = 2, w = 0.86$

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
100	BIAS	alpha	0.0278 ⁽⁸⁾	-0.0056 ⁽³⁾	-0.0030 ⁽⁴⁾	1.8696 ⁽⁹⁾	0.0100 ⁽⁶⁾	0.0178 ⁽⁷⁾	-0.0341 ⁽²⁾	-0.0402 ⁽¹⁾
		beta	0.0563 ⁽²⁾	0.1432 ⁽³⁾	0.1899 ⁽⁶⁾	-1.6382 ⁽¹⁾	0.1619 ⁽⁴⁾	0.1657 ⁽⁵⁾	0.1918 ⁽⁸⁾	0.2221 ⁽⁹⁾
		theta	0.3900 ⁽⁵⁾	0.5433 ⁽⁶⁾	0.9173 ⁽⁸⁾	-0.3508 ⁽¹⁾	0.6212 ⁽⁷⁾	0.3342 ⁽³⁾	0.1466 ⁽²⁾	0.3681 ⁽⁴⁾
	MSE	alpha	0.1656 ⁽¹⁾	0.1755 ⁽²⁾	0.2463 ⁽⁷⁾	3.6395 ⁽⁹⁾	0.2336 ⁽⁶⁾	0.2288 ⁽⁵⁾	0.2068 ⁽⁴⁾	0.1874 ⁽³⁾
		beta	0.3630 ⁽¹⁾	0.5042 ⁽²⁾	0.7360 ⁽⁶⁾	2.6971 ⁽⁹⁾	0.6717 ⁽⁵⁾	0.8119 ⁽⁸⁾	0.6212 ⁽⁴⁾	0.5455 ⁽³⁾
		theta	1.5207 ⁽³⁾	4.4252 ⁽⁶⁾	8.7692 ⁽⁸⁾	0.3125 ⁽¹⁾	4.8364 ⁽⁷⁾	1.4843 ⁽²⁾	1.5560 ⁽⁴⁾	3.0674 ⁽⁵⁾
	MRE	alpha	0.1493 ⁽¹⁾	0.1614 ⁽³⁾	0.1894 ⁽⁷⁾	0.9348 ⁽⁹⁾	0.1827 ⁽⁵⁾	0.1861 ⁽⁶⁾	0.1764 ⁽⁴⁾	0.1599 ⁽²⁾
		beta	0.1592 ⁽¹⁾	0.1819 ⁽²⁾	0.2161 ⁽⁶⁾	0.5461 ⁽⁹⁾	0.2093 ⁽⁵⁾	0.2218 ⁽⁸⁾	0.2057 ⁽⁴⁾	0.1883 ⁽³⁾
		theta	0.3668 ⁽²⁾	0.4820 ⁽⁶⁾	0.7088 ⁽⁸⁾	0.2251 ⁽¹⁾	0.5400 ⁽⁷⁾	0.3869 ⁽⁴⁾	0.3807 ⁽³⁾	0.4461 ⁽⁵⁾
	$\sum$ Ranks		24	33	60	49	52	48	35	35
150	BIAS	alpha	-0.0392 ⁽⁷⁾	-0.0542 ⁽⁶⁾	-0.0694 ⁽³⁾	1.6656 ⁽⁹⁾	-0.0673 ⁽⁵⁾	-0.0243 ⁽⁸⁾	-0.1071 ⁽¹⁾	-0.0945 ⁽²⁾
		beta	0.0999 ⁽²⁾	0.1427 ⁽⁴⁾	0.1926 ⁽⁶⁾	-1.5388 ⁽¹⁾	0.1886 ⁽⁵⁾	0.1172 ⁽³⁾	0.2470 ⁽⁹⁾	0.2351 ⁽⁸⁾
		theta	0.1614 ⁽³⁾	0.1640 ⁽⁴⁾	0.5225 ⁽⁸⁾	-0.5078 ⁽¹⁾	0.5029 ⁽⁷⁾	0.0905 ⁽²⁾	0.2048 ⁽⁵⁾	0.2533 ⁽⁶⁾
	MSE	alpha	0.0801 ⁽¹⁾	0.0878 ⁽²⁾	0.1106 ⁽⁵⁾	2.9846 ⁽⁹⁾	0.1102 ⁽⁴⁾	0.1163 ⁽⁷⁾	0.1394 ⁽⁸⁾	0.1039 ⁽³⁾
		beta	0.2364 ⁽¹⁾	0.2880 ⁽²⁾	0.4428 ⁽⁶⁾	2.4163 ⁽⁹⁾	0.4384 ⁽⁵⁾	0.4029 ⁽⁴⁾	0.4580 ⁽⁸⁾	0.3817 ⁽³⁾
		theta	0.2884 ⁽¹⁾	0.3682 ⁽³⁾	4.2977 ⁽⁸⁾	0.4106 ⁽⁴⁾	3.6891 ⁽⁶⁾	0.2977 ⁽²⁾	1.3206 ⁽⁵⁾	3.7448 ⁽⁷⁾
	MRE	alpha	0.1128 ⁽¹⁾	0.1184 ⁽²⁾	0.1326 ⁽⁵⁾	0.8328 ⁽⁹⁾	0.1315 ⁽⁴⁾	0.1426 ⁽⁸⁾	0.1385 ⁽⁷⁾	0.1229 ⁽³⁾
		beta	0.1302 ⁽¹⁾	0.1432 ⁽²⁾	0.1664 ⁽⁵⁾	0.5129 ⁽⁹⁾	0.1660 ⁽⁴⁾	0.1684 ⁽⁷⁾	0.1708 ⁽⁸⁾	0.1554 ⁽³⁾
		theta	0.2002 ⁽¹⁾	0.2255 ⁽³⁾	0.4372 ⁽⁸⁾	0.2822 ⁽⁴⁾	0.4285 ⁽⁷⁾	0.2004 ⁽²⁾	0.2939 ⁽⁵⁾	0.3107 ⁽⁶⁾
	$\sum$ Ranks		18	28	54	55	47	43	56	41
200	BIAS	alpha	0.0285 ⁽⁷⁾	0.0187 ⁽⁵⁾	-0.0003 ⁽²⁾	1.5062 ⁽⁹⁾	0.0138 ⁽⁴⁾	0.0546 ⁽⁸⁾	0.0242 ⁽⁶⁾	-0.0069 ⁽¹⁾
		beta	-0.0076 ⁽³⁾	0.0133 ⁽⁴⁾	0.0469 ⁽⁶⁾	-1.4249 ⁽¹⁾	0.0229 ⁽⁵⁾	-0.0332 ⁽²⁾	0.0632 ⁽⁸⁾	0.0701 ⁽⁹⁾
		theta	0.0250 ⁽⁵⁾	0.1584 ⁽⁶⁾	0.4521 ⁽⁸⁾	-0.4854 ⁽¹⁾	0.3710 ⁽⁷⁾	-0.0415 ⁽³⁾	-0.0336 ⁽⁴⁾	-0.0814 ⁽²⁾
	MSE	alpha	0.0694 ⁽¹⁾	0.0731 ⁽²⁾	0.0867 ⁽⁵⁾	2.5627 ⁽⁹⁾	0.0864 ⁽⁴⁾	0.0951 ⁽⁷⁾	0.1245 ⁽⁸⁾	0.0731 ⁽²⁾
		beta	0.1518 ⁽¹⁾	0.1809 ⁽²⁾	0.2126 ⁽⁵⁾	2.1326 ⁽⁹⁾	0.2071 ⁽⁴⁾	0.2248 ⁽⁷⁾	0.3171 ⁽⁸⁾	0.1853 ⁽³⁾
		theta	0.1231 ⁽¹⁾	2.7667 ⁽⁶⁾	4.8020 ⁽⁸⁾	0.5661 ⁽⁵⁾	3.6169 ⁽⁷⁾	0.1269 ⁽²⁾	0.3110 ⁽⁴⁾	0.1339 ⁽³⁾
	MRE	alpha	0.1060 ⁽¹⁾	0.1110 ⁽³⁾	0.1191 ⁽⁵⁾	0.7531 ⁽⁹⁾	0.1162 ⁽⁴⁾	0.1220 ⁽⁷⁾	0.1341 ⁽⁸⁾	0.1099 ⁽²⁾
		beta								
		theta								
	$\sum$ Ranks									

Table 5 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
250	$\sum$ Ranks	beta	0.1038 ⁽¹⁾	0.1124 ⁽²⁾	0.1215 ⁽⁵⁾	0.4750 ⁽⁹⁾	0.1191 ⁽⁴⁾	0.1297 ⁽⁷⁾	0.1444 ⁽⁸⁾	0.1134 ⁽³⁾
		theta	0.1385 ⁽¹⁾	0.2375 ⁽⁵⁾	0.4149 ⁽⁸⁾	0.3325 ⁽⁶⁾	0.3686 ⁽⁷⁾	0.1483 ⁽²⁾	0.2149 ⁽⁴⁾	0.1505 ⁽³⁾
	BIAS		21	35	52	58	46	45	58	28
		alpha	0.0161 ⁽⁴⁾	0.0182 ⁽⁵⁾	0.0360 ⁽⁶⁾	1.3323 ⁽⁹⁾	0.0427 ⁽⁸⁾	0.0082 ⁽³⁾	-0.0412 ⁽¹⁾	-0.0114 ⁽²⁾
	MSE	beta	0.0006 ⁽⁵⁾	0.0055 ⁽⁶⁾	-0.0093 ⁽³⁾	-1.3162 ⁽¹⁾	-0.0180 ⁽²⁾	0.0560 ⁽⁷⁾	0.1132 ⁽⁹⁾	0.0601 ⁽⁸⁾
		theta	0.0941 ⁽⁹⁾	0.0456 ⁽⁶⁾	0.0230 ⁽⁴⁾	-0.5273 ⁽¹⁾	0.0109 ⁽³⁾	0.0853 ⁽⁸⁾	0.0786 ⁽⁷⁾	0.0045 ⁽²⁾
	MRE	alpha	0.0412 ⁽¹⁾	0.0485 ⁽³⁾	0.0651 ⁽⁴⁾	2.0460 ⁽⁹⁾	0.0683 ⁽⁶⁾	0.0774 ⁽⁸⁾	0.0711 ⁽⁷⁾	0.0430 ⁽²⁾
		beta	0.1195 ⁽¹⁾	0.1496 ⁽³⁾	0.1977 ⁽⁴⁾	1.8626 ⁽⁹⁾	0.2089 ⁽⁶⁾	0.2798 ⁽⁸⁾	0.2679 ⁽⁷⁾	0.1391 ⁽²⁾
		theta	0.1427 ⁽²⁾	0.1715 ⁽³⁾	0.2357 ⁽⁶⁾	0.5398 ⁽⁹⁾	0.2311 ⁽⁵⁾	0.1998 ⁽⁴⁾	0.2365 ⁽⁸⁾	0.1415 ⁽¹⁾
		alpha	0.0857 ⁽¹⁾	0.0919 ⁽³⁾	0.1046 ⁽⁴⁾	0.6661 ⁽⁹⁾	0.1063 ⁽⁶⁾	0.1128 ⁽⁷⁾	0.1129 ⁽⁸⁾	0.0869 ⁽²⁾
	$\sum$ Ranks	beta	0.0955 ⁽¹⁾	0.1055 ⁽³⁾	0.1184 ⁽⁴⁾	0.4387 ⁽⁹⁾	0.1199 ⁽⁶⁾	0.1402 ⁽⁸⁾	0.1365 ⁽⁷⁾	0.1022 ⁽²⁾
		theta	0.1396 ⁽¹⁾	0.1418 ⁽²⁾	0.1947 ⁽⁶⁾	0.4159 ⁽⁹⁾	0.1962 ⁽⁸⁾	0.1553 ⁽⁴⁾	0.1827 ⁽⁵⁾	0.1439 ⁽³⁾
300	BIAS		25	35	42	65	48	57	59	23
		alpha	-0.0144 ⁽²⁾	-0.0138 ⁽³⁾	-0.0130 ⁽⁴⁾	1.1582 ⁽⁹⁾	-0.0128 ⁽⁶⁾	-0.0091 ⁽⁷⁾	-0.0040 ⁽⁸⁾	-0.0366 ⁽¹⁾
	MSE	beta	0.0572 ⁽²⁾	0.0588 ⁽³⁾	0.0708 ⁽⁶⁾	-1.1784 ⁽¹⁾	0.0686 ⁽⁵⁾	0.0798 ⁽⁸⁾	0.0678 ⁽⁴⁾	0.1089 ⁽⁹⁾
		theta	0.0559 ⁽⁶⁾	0.0210 ⁽⁴⁾	0.0693 ⁽⁷⁾	-0.3099 ⁽¹⁾	0.0719 ⁽⁹⁾	0.0421 ⁽⁵⁾	0.0088 ⁽³⁾	-0.0199 ⁽²⁾
	MRE	alpha	0.0444 ⁽²⁾	0.0441 ⁽¹⁾	0.0579 ⁽⁴⁾	1.6812 ⁽⁹⁾	0.0580 ⁽⁶⁾	0.0738 ⁽⁷⁾	0.0789 ⁽⁸⁾	0.0496 ⁽³⁾
		beta	0.1204 ⁽¹⁾	0.1342 ⁽²⁾	0.1813 ⁽⁵⁾	1.6418 ⁽⁹⁾	0.1779 ⁽⁴⁾	0.2355 ⁽⁸⁾	0.2250 ⁽⁷⁾	0.1478 ⁽³⁾
		theta	0.1237 ⁽¹⁾	0.1278 ⁽³⁾	0.2666 ⁽⁷⁾	3.7923 ⁽⁹⁾	0.2661 ⁽⁶⁾	0.1525 ⁽⁴⁾	0.2361 ⁽⁵⁾	0.1265 ⁽²⁾
		alpha	0.0733 ⁽¹⁾	0.0762 ⁽³⁾	0.0915 ⁽⁴⁾	0.4867 ⁽⁹⁾	0.0924 ⁽⁶⁾	0.0959 ⁽⁷⁾	0.0973 ⁽⁸⁾	0.0758 ⁽²⁾
	$\sum$ Ranks	beta	0.0827 ⁽¹⁾	0.0871 ⁽²⁾	0.1047 ⁽⁶⁾	0.3610 ⁽⁹⁾	0.1044 ⁽⁵⁾	0.1106 ⁽⁸⁾	0.1024 ⁽⁴⁾	0.0887 ⁽³⁾
		theta	0.1095 ⁽¹⁾	0.1220 ⁽⁴⁾	0.2851 ⁽⁶⁾	0.3563 ⁽⁹⁾	0.2885 ⁽⁸⁾	0.1176 ⁽³⁾	0.1491 ⁽⁵⁾	0.1101 ⁽²⁾
	BIAS		22	32	50	62	59	47	49	25
		alpha	-0.0133 ⁽⁶⁾	-0.0146 ⁽⁵⁾	-0.0216 ⁽³⁾	0.9461 ⁽⁹⁾	-0.0230 ⁽²⁾	0.0076 ⁽⁸⁾	-0.0122 ⁽⁷⁾	-0.0325 ⁽¹⁾
350	MSE	beta	0.0201 ⁽³⁾	0.0238 ⁽⁴⁾	0.0468 ⁽⁶⁾	-1.0377 ⁽¹⁾	0.0492 ⁽⁸⁾	0.0020 ⁽²⁾	0.0368 ⁽⁵⁾	0.0624 ⁽⁹⁾
		theta	0.0263 ⁽⁶⁾	0.0159 ⁽⁵⁾	0.2726 ⁽⁷⁾	-0.1831 ⁽¹⁾	0.2893 ⁽⁹⁾	-0.0200 ⁽⁴⁾	-0.0292 ⁽³⁾	-0.0458 ⁽²⁾
		alpha	0.0327 ⁽¹⁾	0.0351 ⁽³⁾	0.0543 ⁽⁵⁾	1.2511 ⁽⁹⁾	0.0551 ⁽⁷⁾	0.0537 ⁽⁴⁾	0.0557 ⁽⁸⁾	0.0346 ⁽²⁾

Table 5 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
MRE		beta	0.0864 ⁽¹⁾	0.0908 ⁽²⁾	0.1528 ⁽⁵⁾	1.3553 ⁽⁹⁾	0.1564 ⁽⁷⁾	0.1580 ⁽⁸⁾	0.1347 ⁽⁴⁾	0.0988 ⁽³⁾
		theta	0.0743 ⁽²⁾	0.1137 ⁽⁴⁾	3.5471 ⁽⁸⁾	2.8181 ⁽⁶⁾	3.4317 ⁽⁷⁾	0.0829 ⁽³⁾	0.1418 ⁽⁵⁾	0.0739 ⁽¹⁾
		alpha	0.0733 ⁽¹⁾	0.0762 ⁽³⁾	0.0915 ⁽⁴⁾	0.4867 ⁽⁹⁾	0.0924 ⁽⁶⁾	0.0959 ⁽⁷⁾	0.0973 ⁽⁸⁾	0.0758 ⁽²⁾
		beta	0.0827 ⁽¹⁾	0.0871 ⁽²⁾	0.1047 ⁽⁶⁾	0.3610 ⁽⁹⁾	0.1044 ⁽⁵⁾	0.1106 ⁽⁸⁾	0.1024 ⁽⁴⁾	0.0887 ⁽³⁾
		theta	0.1095 ⁽¹⁾	0.1220 ⁽⁴⁾	0.2851 ⁽⁶⁾	0.3563 ⁽⁹⁾	0.2885 ⁽⁸⁾	0.1176 ⁽³⁾	0.1491 ⁽⁵⁾	0.1101 ⁽²⁾
$\sum$ Ranks			22	32	50	62	59	47	49	25

Table 6 Simulation values of BIAS, MSE, and MRE for $\alpha = 4$, $\beta = 2$, $\theta = 1$, $w = 0.9$

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
100	BIAS	alpha	0.1181 ⁽⁸⁾	0.0906 ⁽⁶⁾	0.0831 ⁽³⁾	1.4583 ⁽⁹⁾	0.0884 ⁽⁵⁾	0.1123 ⁽⁷⁾	-0.0682 ⁽¹⁾	-0.0317 ⁽²⁾
		beta	0.0042 ⁽²⁾	0.0363 ⁽³⁾	0.0490 ⁽⁶⁾	-0.6095 ⁽¹⁾	0.0486 ⁽⁵⁾	0.0435 ⁽⁴⁾	0.1186 ⁽⁹⁾	0.1049 ⁽⁸⁾
	MSE	theta	0.1964 ⁽⁵⁾	0.1937 ⁽⁴⁾	1.1096 ⁽⁸⁾	0.9216 ⁽⁷⁾	0.6360 ⁽⁶⁾	0.1302 ⁽³⁾	0.0676 ⁽²⁾	0.0301 ⁽¹⁾
		alpha	0.5986 ⁽¹⁾	0.7186 ⁽⁴⁾	0.8577 ⁽⁶⁾	2.4329 ⁽⁹⁾	0.8132 ⁽⁵⁾	0.8852 ⁽⁸⁾	0.6960 ⁽³⁾	0.6714 ⁽²⁾
	MRE	beta	0.1262 ⁽¹⁾	0.1635 ⁽²⁾	0.2087 ⁽⁶⁾	0.3946 ⁽⁹⁾	0.2077 ⁽⁵⁾	0.2213 ⁽⁸⁾	0.2073 ⁽⁴⁾	0.1697 ⁽³⁾
		theta	0.2989 ⁽⁴⁾	0.3560 ⁽⁵⁾	15.6284 ⁽⁸⁾	3.5022 ⁽⁶⁾	4.9419 ⁽⁷⁾	0.2776 ⁽³⁾	0.2501 ⁽²⁾	0.2024 ⁽¹⁾
150	BIAS	alpha	0.1325 ⁽¹⁾	0.1520 ⁽³⁾	0.1717 ⁽⁶⁾	0.3666 ⁽⁹⁾	0.1672 ⁽⁵⁾	0.1742 ⁽⁸⁾	0.1532 ⁽⁴⁾	0.1395 ⁽²⁾
		beta	0.1365 ⁽¹⁾	0.1542 ⁽²⁾	0.1761 ⁽⁵⁾	0.3047 ⁽⁹⁾	0.1773 ⁽⁷⁾	0.1812 ⁽⁸⁾	0.1745 ⁽⁴⁾	0.1603 ⁽³⁾
	MSE	theta	0.3550 ⁽⁴⁾	0.3756 ⁽⁵⁾	1.3229 ⁽⁷⁾	1.3380 ⁽⁹⁾	0.8260 ⁽⁶⁾	0.3344 ⁽²⁾	0.3377 ⁽³⁾	0.3033 ⁽¹⁾
			27	34	55	68	51	51	32	23
	MRE	alpha	0.0596 ⁽⁷⁾	0.0123 ⁽⁶⁾	-0.0004 ⁽²⁾	1.4559 ⁽⁹⁾	0.0096 ⁽⁵⁾	0.0830 ⁽⁸⁾	0.0080 ⁽⁴⁾	-0.0348 ⁽¹⁾
		beta	0.0230 ⁽²⁾	0.0495 ⁽⁴⁾	0.0610 ⁽⁶⁾	-0.5734 ⁽¹⁾	0.0587 ⁽⁵⁾	0.0399 ⁽³⁾	0.0826 ⁽⁸⁾	0.0877 ⁽⁹⁾
200	BIAS	theta	0.0205 ⁽⁵⁾	0.0049 ⁽⁴⁾	0.6219 ⁽⁷⁾	0.6390 ⁽⁹⁾	0.5506 ⁽⁶⁾	-0.0418 ⁽³⁾	-0.0430 ⁽²⁾	-0.0851 ⁽¹⁾
		alpha	0.3423 ⁽¹⁾	0.3779 ⁽²⁾	0.4754 ⁽⁴⁾	2.8990 ⁽⁹⁾	0.4847 ⁽⁶⁾	0.6494 ⁽⁸⁾	0.5977 ⁽⁷⁾	0.4053 ⁽³⁾
	MSE	beta	0.0823 ⁽¹⁾	0.0953 ⁽²⁾	0.1258 ⁽⁴⁾	0.3736 ⁽⁹⁾	0.1314 ⁽⁶⁾	0.1633 ⁽⁷⁾	0.1735 ⁽⁸⁾	0.1127 ⁽³⁾
		theta	0.0655 ⁽³⁾	0.1072 ⁽⁵⁾	7.5054 ⁽⁸⁾	2.1015 ⁽⁶⁾	7.1966 ⁽⁷⁾	0.0655 ⁽³⁾	0.0654 ⁽²⁾	0.0630 ⁽¹⁾
	MRE	alpha	0.1129 ⁽¹⁾	0.1141 ⁽²⁾	0.1310 ⁽⁴⁾	0.3686 ⁽⁹⁾	0.1331 ⁽⁶⁾	0.1560 ⁽⁸⁾	0.1441 ⁽⁷⁾	0.1216 ⁽³⁾
		beta	0.1113 ⁽¹⁾	0.1159 ⁽²⁾	0.1331 ⁽⁴⁾	0.2916 ⁽⁹⁾	0.1361 ⁽⁶⁾	0.1606 ⁽⁸⁾	0.1568 ⁽⁷⁾	0.1265 ⁽³⁾
200	BIAS	theta	0.1823 ⁽²⁾	0.2272 ⁽⁵⁾	0.9090 ⁽⁷⁾	1.0850 ⁽⁹⁾	0.8182 ⁽⁶⁾	0.1787 ⁽¹⁾	0.2012 ⁽⁴⁾	0.1964 ⁽³⁾
			23	32	46	70	53	49	49	27
	MSE	alpha	0.0431 ⁽⁷⁾	0.0394 ⁽⁶⁾	0.0288 ⁽⁴⁾	1.1280 ⁽⁹⁾	0.0251 ⁽³⁾	0.0773 ⁽⁸⁾	-0.0929 ⁽¹⁾	-0.0060 ⁽²⁾
		beta	0.0012 ⁽³⁾	0.0084 ⁽⁴⁾	0.0185 ⁽⁵⁾	-0.4595 ⁽¹⁾	0.0226 ⁽⁷⁾	-0.0015 ⁽²⁾	0.1065 ⁽⁹⁾	0.0355 ⁽⁸⁾
	MRE	theta	0.0665 ⁽⁴⁾	0.0931 ⁽⁵⁾	0.4772 ⁽⁷⁾	0.7568 ⁽⁹⁾	0.4447 ⁽⁶⁾	0.0358 ⁽³⁾	0.0002 ⁽²⁾	-0.0264 ⁽¹⁾
		alpha	0.2004 ⁽¹⁾	0.2421 ⁽³⁾	0.3004 ⁽⁴⁾	2.0244 ⁽⁹⁾	0.3169 ⁽⁶⁾	0.3386 ⁽⁷⁾	0.3942 ⁽⁸⁾	0.2340 ⁽²⁾
200	BIAS	beta	0.0489 ⁽¹⁾	0.0617 ⁽²⁾	0.0854 ⁽⁴⁾	0.2800 ⁽⁹⁾	0.0893 ⁽⁶⁾	0.0909 ⁽⁷⁾	0.1305 ⁽⁸⁾	0.0653 ⁽³⁾
		theta	0.0667 ⁽⁴⁾	0.1570 ⁽⁵⁾	4.7603 ⁽⁸⁾	2.3137 ⁽⁶⁾	2.8773 ⁽⁷⁾	0.0592 ⁽²⁾	0.0621 ⁽³⁾	0.0514 ⁽¹⁾
	MRE	alpha	0.0840 ⁽¹⁾	0.0952 ⁽²⁾	0.1090 ⁽⁵⁾	0.2942 ⁽⁹⁾	0.1141 ⁽⁷⁾	0.1071 ⁽⁴⁾	0.1244 ⁽⁸⁾	0.0955 ⁽³⁾
	MRE	beta								
		theta								

Table 6 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
250	$\sum$ Ranks	beta	0.0890 ⁽¹⁾	0.1011 ⁽²⁾	0.1180 ⁽⁵⁾	0.2456 ⁽⁹⁾	0.1226 ⁽⁷⁾	0.1119 ⁽⁴⁾	0.1312 ⁽⁸⁾	0.1036 ⁽³⁾
		theta	0.1957 ⁽⁴⁾	0.2692 ⁽⁵⁾	0.6908 ⁽⁷⁾	0.6908 ⁽⁷⁾	1.0766 ⁽⁹⁾	0.6374 ⁽⁶⁾	0.1911 ⁽²⁾	0.1955 ⁽³⁾
	BIAS	alpha	26	34	49	70	55	39	50	24
			−0.0004 ⁽⁴⁾	−0.0066 ⁽³⁾	0.0039 ⁽⁵⁾	1.0373 ⁽⁹⁾	0.0235 ⁽⁸⁾	0.0229 ⁽⁷⁾	−0.0097 ⁽²⁾	−0.0498 ⁽¹⁾
	MSE	beta	0.0327 ⁽²⁾	0.0438 ⁽⁴⁾	0.0441 ⁽⁵⁾	−0.4107 ⁽¹⁾	0.0392 ⁽³⁾	0.0474 ⁽⁷⁾	0.0631 ⁽⁸⁾	0.0637 ⁽⁹⁾
		theta	0.0383 ⁽⁴⁾	0.0462 ⁽⁵⁾	0.1248 ⁽⁷⁾	0.3612 ⁽⁹⁾	0.1241 ⁽⁶⁾	0.136 ⁽³⁾	−0.0138 ⁽²⁾	−0.0376 ⁽¹⁾
	MRE	alpha	0.1806 ⁽¹⁾	0.2418 ⁽³⁾	0.2990 ⁽⁴⁾	1.7733 ⁽⁹⁾	0.3425 ⁽⁶⁾	0.3824 ⁽⁸⁾	0.3523 ⁽⁷⁾	0.1958 ⁽²⁾
		beta	0.0487 ⁽¹⁾	0.0645 ⁽³⁾	0.0781 ⁽⁴⁾	0.2501 ⁽⁹⁾	0.0849 ⁽⁶⁾	0.1076 ⁽⁸⁾	0.1026 ⁽⁷⁾	0.0582 ⁽²⁾
	MRE	theta	0.0490 ⁽²⁾	0.0882 ⁽⁵⁾	0.2363 ⁽⁷⁾	2.0356 ⁽⁹⁾	0.2290 ⁽⁶⁾	0.0502 ⁽³⁾	0.0776 ⁽⁴⁾	0.0407 ⁽¹⁾
		alpha	0.0878 ⁽¹⁾	0.1033 ⁽³⁾	0.1129 ⁽⁴⁾	0.2774 ⁽⁹⁾	0.1179 ⁽⁶⁾	0.1253 ⁽⁸⁾	0.1222 ⁽⁷⁾	0.0914 ⁽²⁾
	$\sum$ Ranks	beta	0.0891 ⁽¹⁾	0.1059 ⁽³⁾	0.1174 ⁽⁴⁾	0.2257 ⁽⁹⁾	0.1217 ⁽⁶⁾	0.1315 ⁽⁸⁾	0.1271 ⁽⁷⁾	0.0967 ⁽²⁾
		theta	0.1624 ⁽¹⁾	0.2111 ⁽⁵⁾	0.3207 ⁽⁷⁾	0.7976 ⁽⁹⁾	0.3090 ⁽⁶⁾	0.1731 ⁽³⁾	0.1883 ⁽⁴⁾	0.1625 ⁽²⁾
300	$\sum$ Ranks	alpha	17	34	47	73	53	55	48	22
		beta	0.0750 ⁽⁸⁾	0.0289 ⁽⁵⁾	0.0260 ⁽²⁾	1.1489 ⁽⁹⁾	0.0128 ⁽¹⁾	0.0316 ⁽⁷⁾	0.0312 ⁽⁶⁾	0.0282 ⁽⁴⁾
	BIAS	theta	−0.0162 ⁽²⁾	0.0076 ⁽³⁾	0.0096 ⁽⁴⁾	−0.4174 ⁽¹⁾	0.0236 ⁽⁸⁾	0.0235 ⁽⁷⁾	0.0237 ⁽⁹⁾	0.0119 ⁽⁶⁾
		MSE	0.0055 ⁽⁴⁾	0.2150 ⁽⁸⁾	0.2150 ⁽⁸⁾	0.1271 ⁽⁶⁾	0.2140 ⁽⁷⁾	−0.0048 ⁽³⁾	−0.0218 ⁽²⁾	−0.0409 ⁽¹⁾
	MRE	alpha	0.1700 ⁽¹⁾	0.1971 ⁽³⁾	0.2244 ⁽⁴⁾	2.5763 ⁽⁹⁾	0.2517 ⁽⁶⁾	0.3054 ⁽⁷⁾	0.3287 ⁽⁸⁾	0.1740 ⁽²⁾
		beta	0.0431 ⁽¹⁾	0.0523 ⁽³⁾	0.0578 ⁽⁴⁾	0.2970 ⁽⁹⁾	0.0732 ⁽⁶⁾	0.0993 ⁽⁸⁾	0.0783 ⁽⁷⁾	0.0489 ⁽²⁾
	MRE	theta	0.0435 ⁽²⁾	0.0844 ⁽⁵⁾	3.6327 ⁽⁸⁾	1.2457 ⁽⁶⁾	2.6357 ⁽⁷⁾	0.0495 ⁽³⁾	0.0620 ⁽⁴⁾	0.0393 ⁽¹⁾
		alpha	0.0858 ⁽¹⁾	0.0904 ⁽³⁾	0.0945 ⁽⁴⁾	0.3142 ⁽⁹⁾	0.0970 ⁽⁶⁾	0.1103 ⁽⁸⁾	0.1094 ⁽⁷⁾	0.0865 ⁽²⁾
	$\sum$ Ranks	beta	0.0860 ⁽¹⁾	0.0912 ⁽³⁾	0.0948 ⁽⁴⁾	0.2356 ⁽⁹⁾	0.1009 ⁽⁶⁾	0.1189 ⁽⁸⁾	0.1102 ⁽⁷⁾	0.0904 ⁽²⁾
		theta	0.1540 ⁽¹⁾	0.1989 ⁽⁵⁾	0.4619 ⁽⁷⁾	0.7215 ⁽⁹⁾	0.4339 ⁽⁶⁾	0.1643 ⁽³⁾	0.1834 ⁽⁴⁾	0.1576 ⁽²⁾
	BIAS	alpha	22	34	45	73	57	52	51	23
		beta	0.0700 ⁽³⁾	0.0856 ⁽⁴⁾	0.1091 ⁽⁶⁾	0.9965 ⁽⁹⁾	0.1084 ⁽⁵⁾	0.1149 ⁽⁸⁾	0.0047 ⁽¹⁾	0.0414 ⁽²⁾
350	$\sum$ Ranks	theta	−0.0164 ⁽⁷⁾	−0.0224 ⁽⁶⁾	−0.0313 ⁽²⁾	−0.3818 ⁽¹⁾	−0.0287 ⁽⁴⁾	−0.0275 ⁽⁵⁾	0.0379 ⁽⁹⁾	0.0019 ⁽⁸⁾
		MSE	0.0642 ⁽⁴⁾	0.0678 ⁽⁵⁾	0.0947 ⁽⁶⁾	0.2476 ⁽⁹⁾	0.1077 ⁽⁸⁾	0.0511 ⁽³⁾	0.0240 ⁽²⁾	0.0045 ⁽¹⁾
	MSE	alpha	0.1133 ⁽¹⁾	0.1354 ⁽³⁾	0.1701 ⁽⁴⁾	1.7820 ⁽⁹⁾	0.1819 ⁽⁶⁾	0.2159 ⁽⁸⁾	0.2044 ⁽⁷⁾	0.1305 ⁽²⁾

Table 6 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
MRE		beta	0.0292 ⁽¹⁾	0.0349 ⁽³⁾	0.0439 ⁽⁴⁾	0.2393 ⁽⁹⁾	0.0476 ⁽⁶⁾	0.0611 ⁽⁸⁾	0.0592 ⁽⁷⁾	0.0347 ⁽²⁾
		theta	0.0304 ⁽³⁾	0.0370 ⁽⁴⁾	0.0735 ⁽⁶⁾	1.1069 ⁽⁹⁾	0.0828 ⁽⁸⁾	0.0295 ⁽²⁾	0.0456 ⁽⁵⁾	0.0248 ⁽¹⁾
		alpha	0.0680 ⁽¹⁾	0.0716 ⁽²⁾	0.0807 ⁽⁴⁾	0.2663 ⁽⁹⁾	0.0820 ⁽⁶⁾	0.0915 ⁽⁸⁾	0.0911 ⁽⁷⁾	0.0722 ⁽³⁾
		beta	0.0690 ⁽¹⁾	0.0732 ⁽²⁾	0.0834 ⁽⁴⁾	0.2088 ⁽⁹⁾	0.0860 ⁽⁶⁾	0.0960 ⁽⁷⁾	0.0997 ⁽⁸⁾	0.0754 ⁽³⁾
		theta	0.1285 ⁽²⁾	0.1491 ⁽⁴⁾	0.2104 ⁽⁶⁾	0.7332 ⁽⁹⁾	0.2107 ⁽⁸⁾	0.1330 ⁽³⁾	0.1611 ⁽⁵⁾	0.1236 ⁽¹⁾
$\sum$ Ranks			23	33	42	73	57	52	51	23

Table 7 Simulation values of BIAS, MSE, and MRE for $\alpha = 2.5, \beta = 1, \theta = 1, w = 0.71$

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
100	BIAS	alpha	-0.1143 ⁽¹⁾	0.0890 ⁽³⁾	0.1773 ⁽⁸⁾	0.0539 ⁽²⁾	0.1596 ⁽⁷⁾	0.1533 ⁽⁶⁾	0.1031 ⁽⁵⁾	0.0966 ⁽⁴⁾
		beta	0.0877 ⁽⁸⁾	-0.0063 ⁽⁶⁾	-0.0106 ⁽⁴⁾	0.7879 ⁽⁹⁾	-0.0027 ⁽⁷⁾	-0.0325 ⁽²⁾	-0.0359 ⁽¹⁾	-0.0188 ⁽³⁾
	MSE	theta	0.1516 ⁽⁸⁾	0.0554 ⁽⁴⁾	0.1123 ⁽⁶⁾	0.3069 ⁽⁹⁾	0.0934 ⁽⁵⁾	0.0300 ⁽²⁾	0.0313 ⁽³⁾	0.0068 ⁽¹⁾
		alpha	0.1879 ⁽¹⁾	0.2908 ⁽³⁾	0.5702 ⁽⁸⁾	0.5170 ⁽⁶⁾	0.5549 ⁽⁷⁾	0.3791 ⁽⁵⁾	0.3330 ⁽⁴⁾	0.2092 ⁽²⁾
	MRE	beta	0.0628 ⁽³⁾	0.0635 ⁽⁴⁾	0.1181 ⁽⁷⁾	14.4672 ⁽⁹⁾	0.1129 ⁽⁶⁾	0.0776 ⁽⁵⁾	0.0467 ⁽²⁾	0.0421 ⁽¹⁾
		theta	0.2271 ⁽⁵⁾	0.1711 ⁽³⁾	0.4106 ⁽⁷⁾	7.4118 ⁽⁹⁾	0.2941 ⁽⁶⁾	0.0992 ⁽¹⁾	0.1545 ⁽²⁾	0.1909 ⁽⁴⁾
		alpha	0.1392 ⁽¹⁾	0.1765 ⁽⁴⁾	0.2492 ⁽⁸⁾	0.1596 ⁽³⁾	0.2391 ⁽⁷⁾	0.1895 ⁽⁶⁾	0.1800 ⁽⁵⁾	0.1448 ⁽²⁾
		beta	0.1647 ⁽²⁾	0.2058 ⁽⁴⁾	0.2765 ⁽⁷⁾	0.8810 ⁽⁹⁾	0.2695 ⁽⁶⁾	0.2353 ⁽⁵⁾	0.1747 ⁽³⁾	0.1499 ⁽¹⁾
	$\sum$ Ranks	theta	0.2756 ⁽⁵⁾	0.2675 ⁽³⁾	0.3797 ⁽⁷⁾	0.6807 ⁽⁹⁾	0.3476 ⁽⁶⁾	0.2228 ⁽¹⁾	0.2695 ⁽⁴⁾	0.2422 ⁽²⁾
			34	34	62	65	57	33	29	20
150	BIAS	alpha	-0.0277 ⁽²⁾	0.0964 ⁽³⁾	0.1589 ⁽⁷⁾	-0.0915 ⁽¹⁾	0.1584 ⁽⁶⁾	0.1271 ⁽⁴⁾	0.1653 ⁽⁹⁾	0.1402 ⁽⁵⁾
		beta	0.0367 ⁽⁸⁾	-0.0029 ⁽⁷⁾	-0.0100 ⁽³⁾	0.1514 ⁽⁹⁾	-0.0051 ⁽⁶⁾	-0.0076 ⁽⁵⁾	-0.0449 ⁽¹⁾	-0.0405 ⁽²⁾
	MSE	theta	0.0464 ⁽⁹⁾	-0.0088 ⁽⁴⁾	-0.0017 ⁽⁶⁾	-0.0562 ⁽²⁾	0.0132 ⁽⁸⁾	-0.0034 ⁽⁵⁾	-0.0452 ⁽³⁾	-0.0610 ⁽¹⁾
		alpha	0.1390 ⁽¹⁾	0.2246 ⁽⁵⁾	0.4233 ⁽⁷⁾	0.1670 ⁽³⁾	0.4685 ⁽⁹⁾	0.3052 ⁽⁶⁾	0.1913 ⁽⁴⁾	0.1511 ⁽²⁾
	MRE	beta	0.0344 ⁽³⁾	0.0483 ⁽⁴⁾	0.0662 ⁽⁶⁾	0.0788 ⁽⁹⁾	0.0681 ⁽⁸⁾	0.0659 ⁽⁵⁾	0.0324 ⁽²⁾	0.0241 ⁽¹⁾
		theta	0.0553 ⁽³⁾	0.0807 ⁽⁶⁾	0.1273 ⁽⁷⁾	0.0714 ⁽⁵⁾	0.1573 ⁽⁹⁾	0.0463 ⁽²⁾	0.0639 ⁽⁴⁾	0.0396 ⁽¹⁾
		alpha	0.1176 ⁽¹⁾	0.1490 ⁽⁵⁾	0.1954 ⁽⁷⁾	0.1314 ⁽³⁾	0.1997 ⁽⁹⁾	0.1697 ⁽⁶⁾	0.1481 ⁽⁴⁾	0.1244 ⁽²⁾
		beta	0.1310 ⁽²⁾	0.1625 ⁽⁴⁾	0.2020 ⁽⁶⁾	0.1982 ⁽⁵⁾	0.2040 ⁽⁸⁾	0.2098 ⁽⁹⁾	0.1473 ⁽³⁾	0.1248 ⁽¹⁾
	$\sum$ Ranks	theta	0.1756 ⁽³⁾	0.1911 ⁽⁴⁾	0.2494 ⁽⁷⁾	0.2026 ⁽⁶⁾	0.2638 ⁽⁹⁾	0.1598 ⁽¹⁾	0.1968 ⁽⁵⁾	0.1629 ⁽²⁾
			32	42	56	43	72	43	35	17
200	BIAS	alpha	-0.0439 ⁽²⁾	0.0883 ⁽⁵⁾	0.1409 ⁽⁶⁾	-0.2071 ⁽¹⁾	0.1532 ⁽⁹⁾	0.1471 ⁽⁸⁾	0.0660 ⁽³⁾	0.0850 ⁽⁴⁾
		beta	0.0452 ⁽⁸⁾	-0.0013 ⁽⁵⁾	0.0016 ⁽⁶⁾	0.3776 ⁽⁹⁾	-0.0033 ⁽⁴⁾	-0.0213 ⁽¹⁾	-0.0094 ⁽³⁾	-0.0157 ⁽²⁾
	MSE	theta	0.0477 ⁽⁶⁾	-0.0100 ⁽³⁾	0.0567 ⁽⁸⁾	0.0536 ⁽⁷⁾	0.0357 ⁽⁵⁾	-0.0075 ⁽⁴⁾	-0.0336 ⁽²⁾	-0.0383 ⁽¹⁾
		alpha	0.0874 ⁽¹⁾	0.1666 ⁽⁴⁾	0.3608 ⁽⁷⁾	0.2494 ⁽⁵⁾	0.3648 ⁽⁹⁾	0.2553 ⁽⁶⁾	0.1342 ⁽³⁾	0.0998 ⁽²⁾
	MRE	beta	0.0154 ⁽²⁾	0.0411 ⁽⁴⁾	0.0831 ⁽⁶⁾	3.7038 ⁽⁹⁾	0.0841 ⁽⁸⁾	0.0598 ⁽⁵⁾	0.0216 ⁽³⁾	0.0120 ⁽¹⁾
		theta	0.0394 ⁽⁵⁾	0.0352 ⁽²⁾	0.2655 ⁽⁸⁾	0.1105 ⁽⁶⁾	0.1698 ⁽⁷⁾	0.0360 ⁽³⁾	0.0364 ⁽⁴⁾	0.0279 ⁽¹⁾
		alpha	0.0932 ⁽¹⁾	0.1306 ⁽⁴⁾	0.1964 ⁽⁷⁾	0.1535 ⁽⁶⁾	0.1969 ⁽⁹⁾	0.1531 ⁽⁵⁾	0.1145 ⁽³⁾	0.1000 ⁽²⁾
	$\sum$ Ranks									

Table 7 (continued)

n	Est.	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
250	theta	beta	0.0977 ⁽²⁾	0.1586 ⁽⁴⁾	0.2388 ⁽⁶⁾	0.4119 ⁽⁹⁾	0.2394 ⁽⁸⁾	0.1883 ⁽⁵⁾	0.1186 ⁽³⁾	0.0895 ⁽¹⁾
		0.1535 ⁽⁵⁾	0.1504 ⁽²⁾	0.2755 ⁽⁸⁾	0.2755 ⁽⁸⁾	0.2462 ⁽⁶⁾	0.2470 ⁽⁷⁾	0.1515 ⁽³⁾	0.1529 ⁽⁴⁾	0.1378 ⁽¹⁾
	$\sum$ Ranks		32	33	62	58	66	40	28	15
	BIAS	alpha	-0.0846 ⁽²⁾	0.0195 ⁽⁵⁾	0.0166 ⁽³⁾	-0.3664 ⁽¹⁾	0.0199 ⁽⁶⁾	0.0696 ⁽⁸⁾	0.0777 ⁽⁹⁾	0.0348 ⁽⁷⁾
		beta	0.0536 ⁽⁸⁾	0.0081 ⁽⁴⁾	0.0235 ⁽⁵⁾	0.4459 ⁽⁹⁾	0.0254 ⁽⁷⁾	-0.0083 ⁽²⁾	-0.0160 ⁽¹⁾	-0.0044 ⁽³⁾
		theta	0.0231 ⁽⁵⁾	-0.0033 ⁽⁴⁾	0.0811 ⁽⁶⁾	0.3334 ⁽⁹⁾	0.0911 ⁽⁸⁾	-0.0138 ⁽³⁾	-0.0328 ⁽²⁾	-0.0418 ⁽¹⁾
	MSE	alpha	0.0907 ⁽²⁾	0.1075 ⁽³⁾	0.2181 ⁽⁶⁾	0.2914 ⁽⁹⁾	0.2368 ⁽⁸⁾	0.1597 ⁽⁵⁾	0.1210 ⁽⁴⁾	0.0836 ⁽¹⁾
		beta	0.0220 ⁽²⁾	0.0272 ⁽⁴⁾	0.0552 ⁽⁶⁾	3.7325 ⁽⁹⁾	0.0627 ⁽⁸⁾	0.0424 ⁽⁵⁾	0.0242 ⁽³⁾	0.0159 ⁽¹⁾
		theta	0.0270 ⁽²⁾	0.0310 ⁽⁴⁾	0.2304 ⁽⁶⁾	3.7095 ⁽⁹⁾	0.2717 ⁽⁸⁾	0.0319 ⁽⁵⁾	0.0303 ⁽³⁾	0.0254 ⁽¹⁾
	MRE	alpha	0.0930 ⁽²⁾	0.1077 ⁽³⁾	0.1509 ⁽⁶⁾	0.1713 ⁽⁹⁾	0.1554 ⁽⁸⁾	0.1268 ⁽⁵⁾	0.1126 ⁽⁴⁾	0.0863 ⁽¹⁾
300		beta	0.1080 ⁽²⁾	0.1308 ⁽⁴⁾	0.1849 ⁽⁶⁾	0.4625 ⁽⁹⁾	0.1927 ⁽⁸⁾	0.1582 ⁽⁵⁾	0.1250 ⁽³⁾	0.0900 ⁽¹⁾
		theta	0.1343 ⁽²⁾	0.1391 ⁽³⁾	0.2533 ⁽⁶⁾	0.4163 ⁽⁹⁾	0.2606 ⁽⁸⁾	0.1414 ⁽⁴⁾	0.1418 ⁽⁵⁾	0.1337 ⁽¹⁾
	$\sum$ Ranks		27	34	50	73	69	42	34	17
	BIAS	alpha	-0.0390 ⁽²⁾	0.0311 ⁽³⁾	0.0621 ⁽⁸⁾	-0.3578 ⁽¹⁾	0.0607 ⁽⁷⁾	0.0602 ⁽⁶⁾	0.0357 ⁽⁴⁾	0.0458 ⁽⁵⁾
		beta	0.0342 ⁽⁸⁾	0.0019 ⁽⁷⁾	-0.0009 ⁽⁴⁾	0.2504 ⁽⁹⁾	0.0015 ⁽⁶⁾	-0.0061 ⁽²⁾	-0.0051 ⁽³⁾	-0.0131 ⁽¹⁾
		theta	0.0314 ⁽⁸⁾	0.0020 ⁽³⁾	0.0135 ⁽⁵⁾	0.1991 ⁽⁹⁾	0.0215 ⁽⁷⁾	0.0043 ⁽⁴⁾	-0.0131 ⁽²⁾	-0.0238 ⁽¹⁾
	MSE	alpha	0.0560 ⁽¹⁾	0.0912 ⁽⁴⁾	0.1735 ⁽⁶⁾	0.2128 ⁽⁹⁾	0.2001 ⁽⁸⁾	0.1334 ⁽⁵⁾	0.0823 ⁽³⁾	0.0605 ⁽²⁾
		beta	0.0103 ⁽²⁾	0.0269 ⁽⁴⁾	0.0440 ⁽⁶⁾	0.1036 ⁽⁹⁾	0.0472 ⁽⁸⁾	0.0427 ⁽⁵⁾	0.0212 ⁽³⁾	0.0096 ⁽¹⁾
		theta	0.0269 ⁽³⁾	0.0290 ⁽⁴⁾	0.0603 ⁽⁶⁾	0.1299 ⁽⁹⁾	0.0722 ⁽⁸⁾	0.0258 ⁽²⁾	0.0409 ⁽⁵⁾	0.0217 ⁽¹⁾
	MRE	alpha	0.0778 ⁽¹⁾	0.0946 ⁽⁴⁾	0.1247 ⁽⁶⁾	0.1535 ⁽⁹⁾	0.1261 ⁽⁸⁾	0.1113 ⁽⁵⁾	0.0882 ⁽³⁾	0.0784 ⁽²⁾
350		beta	0.0802 ⁽¹⁾	0.1196 ⁽⁴⁾	0.1621 ⁽⁶⁾	0.2514 ⁽⁹⁾	0.1641 ⁽⁸⁾	0.1500 ⁽⁵⁾	0.1019 ⁽³⁾	0.0805 ⁽²⁾
		theta	0.1204 ⁽¹⁾	0.1322 ⁽⁴⁾	0.1807 ⁽⁶⁾	0.2546 ⁽⁹⁾	0.1887 ⁽⁸⁾	0.1299 ⁽³⁾	0.1474 ⁽⁵⁾	0.1207 ⁽²⁾
	$\sum$ Ranks		27	37	53	73	68	37	31	17
	BIAS	alpha	-0.0508 ⁽²⁾	0.0247 ⁽³⁾	0.0475 ⁽⁷⁾	-0.3264 ⁽¹⁾	0.0401 ⁽⁵⁾	0.0442 ⁽⁶⁾	0.0709 ⁽⁹⁾	0.0271 ⁽⁴⁾
		beta	0.0459 ⁽⁸⁾	0.0111 ⁽⁴⁾	0.0154 ⁽⁵⁾	0.2477 ⁽⁹⁾	0.0177 ⁽⁷⁾	0.0081 ⁽³⁾	-0.0121 ⁽¹⁾	0.0043 ⁽²⁾
		theta	0.0351 ⁽⁷⁾	-0.0066 ⁽³⁾	0.0281 ⁽⁵⁾	0.1991 ⁽⁹⁾	0.0424 ⁽⁸⁾	-0.0014 ⁽⁴⁾	-0.0339 ⁽¹⁾	-0.0319 ⁽²⁾
	MSE	alpha	0.0530 ⁽¹⁾	0.0923 ⁽⁴⁾	0.1852 ⁽⁶⁾	0.2198 ⁽⁹⁾	0.1863 ⁽⁸⁾	0.1315 ⁽⁵⁾	0.0858 ⁽³⁾	0.0536 ⁽²⁾

Table 7 (continued)

n	Est	Est. Par	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
MRE		beta	0.0120 ⁽²⁾	0.0200 ⁽⁴⁾	0.0437 ⁽⁶⁾	0.1254 ⁽⁹⁾	0.0451 ⁽⁸⁾	0.0347 ⁽⁵⁾	0.0144 ⁽³⁾	0.0088 ⁽¹⁾
		theta	0.0257 ⁽³⁾	0.0263 ⁽⁴⁾	0.1261 ⁽⁶⁾	0.2117 ⁽⁹⁾	0.1398 ⁽⁸⁾	0.0254 ⁽²⁾	0.0363 ⁽⁵⁾	0.0207 ⁽¹⁾
		alpha	0.0738 ⁽²⁾	0.0936 ⁽³⁾	0.1321 ⁽⁷⁾	0.1472 ⁽⁹⁾	0.1317 ⁽⁶⁾	0.1141 ⁽⁵⁾	0.0983 ⁽⁴⁾	0.0721 ⁽¹⁾
		beta	0.0872 ⁽²⁾	0.1102 ⁽⁴⁾	0.1545 ⁽⁶⁾	0.2500 ⁽⁹⁾	0.1566 ⁽⁸⁾	0.1482 ⁽⁵⁾	0.1000 ⁽³⁾	0.0752 ⁽¹⁾
$\sum$ Ranks		theta	0.1225 ⁽²⁾	0.1307 ⁽⁴⁾	0.2054 ⁽⁶⁾	0.2888 ⁽⁹⁾	0.2108 ⁽⁸⁾	0.1270 ⁽³⁾	0.1534 ⁽⁵⁾	0.1162 ⁽¹⁾
			29	33	54	73	66	38	34	15

$$\hat{\theta}_{\text{ADE}} = \arg \min_{\theta} \text{ADE}(\theta)$$

8.3 Cramér–von Mises estimation (CvME) (EM3)

Let $y_{(1)} < y_{(2)} < \dots < y_{(n)}$ be the ordered sample values, and $F(y; \theta)$ be the theoretical cumulative distribution function depending on parameter θ Anderson and Darling (1952). Then the Cramér–von Mises criterion to minimize is:

$$\text{CvM}(\theta) = \frac{1}{12n} + \sum_{i=1}^n \left(F(y_{(i)}; \theta) - \frac{2i-1}{2n} \right)^2$$

The CvME estimate $\hat{\theta}_{\text{CvME}}$ is obtained by minimizing this expression:

$$\hat{\theta}_{\text{CvME}} = \arg \min_{\theta} \text{CvM}(\theta)$$

8.4 Weighted least squares estimation (WLSE) (EM4)

It is an extension of the least squares method by assigning different weights to each data point, giving more importance to certain observations based on their reliability or variance (Rao and Sethuraman 2001).

$$\text{WLSE}(\theta) = \sum_{i=1}^n w_i \left(F(y_{(i)}; \theta) - \frac{i}{n+1} \right)^2$$

where w_i are the weights (commonly taken as $w_i = \frac{1}{F(y_{(i)}; \theta)(1-F(y_{(i)}; \theta))}$).

The estimate $\hat{\theta}_{\text{WLSE}}$ is obtained by minimizing this expression:

$$\hat{\theta}_{\text{WLSE}} = \arg \min_{\theta} \text{WLSE}(\theta)$$

8.5 Absolute distance estimator (EM5)

It is a method that involves minimizing the sum of absolute deviations between observed and predicted values.

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n \left| F(y_{(i)}; \theta) - \frac{i-0.5}{n} \right|$$

8.6 Absolute-log distance estimation (ALDE) (EM6)

It is a stable parameter estimation method that minimizes the sum of absolute differences between the logarithms of observed and predicted values Barndorff-Nielsen and Cox (1994).

$$ALDE(\theta) = \sum_{i=1}^n \left| \log F(y_{(i)}; \theta) - \log \left(\frac{i}{n+1} \right) \right|$$

The ALDE estimator $\hat{\theta}_{ALDE}$ is obtained by minimizing the above expression:

$$\hat{\theta}_{ALDE} = \arg \min_{\theta} ALDE(\theta)$$

8.7 Minimum spacing absolute distance estimator (MSADE) (EM7)

MSADE is useful in situations where data are ordered or where the goal is to find parameter estimates that maximize data spacings. It is a statistical estimation method that aims to estimate parameters by minimizing the total absolute distances between adjacent data points Cheng and Amin (1983).

$$MSADE(\theta) = \sum_{i=1}^{n-1} \left| F(y_{(i+1)}; \theta) - F(y_{(i)}; \theta) - \frac{1}{n} \right|$$

The estimate $\hat{\theta}_{MSADE}$ is obtained by minimizing this expression:

$$\hat{\theta}_{MSADE} = \arg \min_{\theta} MSADE(\theta)$$

8.8 Minimum spacing absolute log distance estimator (MSALDE) (EM8)

The Minimum Spacing Absolute Log Distance (MSALD) estimator is a statistical technique for parameter estimation, particularly in probability distributions, that minimizes the absolute difference between the empirical distribution function (EDF) and the theoretical cumulative distribution function (cdf), but using the logarithm of the data spacings (differences between ordered data points) instead of the raw distances (Ranneby 1984).

$$D_i(\theta) = F(y_{(i)}; \theta) - F(y_{(i-1)}; \theta), \quad i = 1, 2, \dots, n+1$$

where we set

$$\begin{aligned} y_{(0)} &= -\infty \text{ with } F(y_{(0)}; \theta) = 0 \\ y_{(n+1)} &= +\infty \text{ with } F(y_{(n+1)}; \theta) = 1. \end{aligned}$$

The Minimum Spacing Absolute Log Distance Estimator (MSALDE) is:

$$MSALDE(\theta) = \sum_{i=1}^{n+1} \left| \log D_i(\theta) - \log \frac{1}{n+1} \right|$$

The estimate $\hat{\theta}_{\text{MSALDE}}$ is obtained by minimizing this expression:

$$\hat{\theta}_{\text{MSALDE}} = \arg \min_{\theta} MSALDE(\theta)$$

Ranks for simulation measures of all estimation methods are presented in Table 8. The ranking analysis based on bias, MSE and MRE strongly supports the superiority of the Weighted Least Squares Estimation method. WLSE consistently secures the top rank, reflecting its enhanced accuracy and reduced variability compared to alternative estimators. This behavior can be attributed to the weighting structure, which allows WLSE to effectively account for heteroscedasticity and sampling variability.

Algorithm: Simulation Procedure

1. Specify the true parameter values (α, β) and the sample sizes $n \in \{75, 100, 150, 250, 400, 600\}$.
2. Fix the number of Monte Carlo replications $N = 1000$.
3. For each sample size n and replication $r = 1, 2, \dots, N$:
 - (a) Generate a random sample $X_1, X_2, \dots, X_n$ from the proposed distribution using the inverse transformation method.
 - (b) Estimate the model parameters (α, β) using the following estimation methods:
 - Maximum Likelihood Estimation (MLE),
 - Anderson–Darling Estimation (ADE),
 - Cramér–von Mises Estimation (CvME),
 - Least Squares Estimation (LSE),
 - Weighted Least Squares Estimation (WLSE),
 - Absolute Distance Estimation (AbsDistE),
 - Absolute Log-Distance Estimation (ALDE),
 - Minimum Spacing Absolute Distance Estimation (MSADE),
 - Minimum Spacing Absolute Log-Distance Estimation (MSALDE),
 - Maximum Product of Spacings Estimation (MPS).
 - (c) Store the obtained parameter estimates.
4. After completing the N replications, compute the performance measures:

$$\text{Bias}(\hat{\theta}) = \frac{1}{N} \sum_{r=1}^N (\hat{\theta}_r - \theta),$$

Table 8 Parameter values across different EM

Parameter	EM							
n	EM1	EM2	EM3	EM4	EM5	EM6	EM7	EM8
$\alpha=3, \beta=5, \theta=3, w=0.9375$								
100	3	7	2	8	6	1	3	4
150	4	7	2.5	8	6	2.5	1	5
200	1	7	3	8	6	5	4	2
250	2	7	4	8	6	5	3	1
300	2	7	3	8	6	5	4	1
350	1.5	6	3	8	7	5	4	1.5
$\alpha=3, \beta=2, \theta=3, w=0.86$								
100	1	4	6	8	7	5	3	2
150	1	3	5	7	8	6	4	2
200	1	3	6	8	7	5	4	2
250	1	3	4	8	7	6	5	2
300	1	3	5	8	6	4	7	2
350	1	4	6	7	8	4	4	2
$\alpha=2, \beta=3, \theta=2, w=0.86$								
100	1	2	8	6	7	5	3.5	3.5
150	1	2	6	7	5	4	8	3
200	1	3	6	7.5	5	4	7.5	2
250	2	3	4	8	5	6	7	1
300	1	2	4	8	5	7	6	3
350	1	3	6	8	7	4	5	2
$\alpha=2.5, \beta=1, \theta=1, w=0.71$								
100	4.5	4.5	7	8	6	3	2	1
150	2	4	7	5.5	8	5.5	3	1
200	3	4	7	6	8	5	2	1
250	2	3.5	6	8	7	5	3.5	1
300	2	4.5	6	8	7	4.5	3	1
350	2	3	6	8	7	5	4	1
$\alpha=4, \beta=2, \theta=1, w=0.9$								
100	2	4	7	8	5.5	5.5	3	1
150	1	3	4	8	7	5.5	5.5	2
200	2	3	5	8	7	4	6	1
250	1	3	4	8	6	7	5	2
300	1.5	3	4	8	5	6.5	6.5	1.5
350	1.5	3	4	8	7	6	5	1.5
Sum of ranks	51	119.5	150.5	230	194.5	146	131.5	56
Overall rank	1	3	6	8	7	5	4	2

$$\text{MSE}(\hat{\theta}) = \frac{1}{N} \sum_{r=1}^N \left(\hat{\theta}_r - \theta \right)^2,$$

$$\text{MRE}(\hat{\theta}) = \frac{1}{N} \sum_{r=1}^N \left| \frac{\hat{\theta}_r - \theta}{\theta} \right|.$$

5. Compare the performance of the estimation methods based on the computed Bias, MSE, and MRE values.

Table 8 shows the overall ranks of all the methods of estimation of the proposed distribution by various values of model parameters.

9 Order statistics

Let $Y_{(1)}, Y_{(2)}, Y_{(3)}..$ be the order statistics of a random sample $Y_1, Y_2, Y_3..$ drawn from continuous population with PDF $f_y(y)$ and cdf $F_y(y)$ then pdf of r th order statistics $y_{(r)}$ is given by

$$f_{y_{(r)}}(y) = \frac{n!}{(r-1)!(n-r)!} f_y(y) [F_y(y)]^{r-1} [1 - F_y(y)]^{n-r} \quad (39)$$

$$f_{y_{(r)}}(y) = \frac{n!}{(r-1)!(n-r)!} \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{y^{\alpha-1} e^{-y/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta+1} \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)} \right] \\ \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right]^{r-1} \left[1 - \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right] \right]^{n-r}$$

PDF of higher order statistics y_n can be obtained as

$$f_{y_{(n)}}(y) = n \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{y^{\alpha-1} e^{-y/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta+1} \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)} \right] \\ \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right]^{n-1} \quad (40)$$

PDF of first order statistics y_1 can be obtained as

$$f_{y_{(1)}}(y) = n \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{y^{\alpha-1} e^{-y/\beta}}{\beta^\alpha \Gamma(\alpha)} + \frac{1}{\alpha\beta+1} \frac{y^{\theta-1} (1+y)^{-(\theta+\beta)}}{B(\theta, \beta)} \right] \\ \left[1 - \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma(\alpha, \frac{y}{\beta})}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right] \right]^{n-1} \quad (41)$$

Let $Y_{(1)}, Y_{(2)}, \dots, Y_{(n)}$ denote the order statistics of a random sample $Y_1, Y_2, \dots, Y_n$ from the proposed distribution with cumulative distribution function $F_Y(y)$. The cumulative distribution function (cdf) of the r th order statistic $Y_{(r)}$ is given by

$$F_{Y_{(r)}}(y) = \sum_{j=r}^n \binom{n}{j} [F_Y(y)]^j [1 - F_Y(y)]^{n-j}. \quad (42)$$

Substituting the CDF of the proposed distribution,

$$F_{Y_{(r)}}(y) = \sum_{j=r}^n \binom{n}{j} \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right]^j \times \left[1 - \left(\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right) \right]^{n-j}. \quad (43)$$

The cdf of the largest order statistic y_n is obtained by setting ($r=n$)

$$F_{Y_{(n)}}(y) = \left[\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right]^n. \quad (44)$$

Similarly, the cdf of the smallest order statistic y_1 is obtained by setting ($r=1$), which gives

$$F_{Y_{(1)}}(y) = 1 - \left[1 - \left(\frac{\alpha\beta}{\alpha\beta+1} \frac{\gamma\left(\alpha, \frac{y}{\beta}\right)}{\Gamma(\alpha)} + \frac{1}{\alpha\beta+1} I_{\frac{y}{1+y}}(\theta, \beta) \right) \right]^n. \quad (45)$$

10 Renyi and Tsallis entropy of the gamma–beta prime mixture

Entropy measures quantify the uncertainty, diversity, or disorder inherent in a probability distribution and provide valuable insight into its information content and tail behavior. Shannon entropy represents the classical measure of average uncertainty. Rényi and Tsallis entropies generalize Shannon entropy by introducing order parameters that allow differential emphasis on rare or frequent events, making them particularly useful for analyzing heavy-tailed and complex distributions. In this section, we derive Rényi and Tsallis entropy measures for the proposed GBPD distribution, presenting general expressions and obtaining closed-form results for specific integer orders.

Consider the mixture representation of the GBPD random variable Y given by

$$Y \sim wY_1 + (1-w)Y_2, \quad w = \frac{\alpha\beta}{\alpha\beta+1},$$

where

$$Y_1 \sim \text{Gamma}(\alpha, \beta), \quad Y_2 \sim \text{Beta-prime}(\theta, \beta).$$

Accordingly, the probability density function of Y can be written as

$$f(y) = wf_1(y) + (1 - w)f_2(y),$$

with component densities

$$f_1(y) = \frac{y^{\alpha-1}e^{-y/\beta}}{\beta^\alpha\Gamma(\alpha)}, \quad f_2(y) = \frac{y^{\theta-1}(1+y)^{-(\theta+\beta)}}{B(\theta, \beta)}, \quad y > 0.$$

10.1 Rényi entropy

The Rényi entropy of order $r > 0$, $r \neq 1$ is defined as

$$H_r^{\text{Rényi}}(Y) = \frac{1}{1-r} \log \int_0^\infty f(y)^r dy.$$

Substituting the mixture PDF:

$$H_r^{\text{Rényi}}(Y) = \frac{1}{1-r} \log \int_0^\infty [wf_1(y) + (1-w)f_2(y)]^r dy.$$

Using the multinomial expansion (for integer r) we can write

$$(wf_1 + (1-w)f_2)^r = \sum_{k=0}^r \binom{r}{k} w^k (1-w)^{r-k} f_1(y)^k f_2(y)^{r-k}.$$

Thus,

$$H_r^{\text{Rényi}}(Y) = \frac{1}{1-r} \log \int_0^\infty \sum_{k=0}^r \binom{r}{k} w^k (1-w)^{r-k} f_1(y)^k f_2(y)^{r-k} dy$$

Note: For non-integer r , closed-form expressions are generally unavailable, and numerical integration is required. The Rényi entropy exists provided $\int_0^\infty f(y)^r dy < \infty$. In particular, for the special case $r = 2$, the Rényi entropy admits a closed-form expression, which is derived below.

10.1.1 Rényi entropy of order $r = 2$

For the special case $r = 2$, the Rényi entropy reduces to

$$H_2^{\text{Rényi}}(Y) = -\log \left(\int_0^\infty f(y)^2 dy \right).$$

Using the mixture representation $f(y) = wf_1(y) + (1 - w)f_2(y)$, we obtain

$$f(y)^2 = w^2 f_1(y)^2 + (1 - w)^2 f_2(y)^2 + 2w(1 - w)f_1(y)f_2(y).$$

Hence,

$$\int_0^\infty f(y)^2 dy = w^2 \int_0^\infty f_1(y)^2 dy + (1 - w)^2 \int_0^\infty f_2(y)^2 dy + 2w(1 - w) \int_0^\infty f_1(y)f_2(y) dy.$$

The first integral is

$$\begin{aligned} \int_0^\infty f_1(y)^2 dy &= \frac{1}{\beta^{2\alpha}\Gamma(\alpha)^2} \int_0^\infty y^{2\alpha-2} e^{-2y/\beta} dy \\ &= \frac{\Gamma(2\alpha - 1)}{2^{2\alpha-1}\beta\Gamma(\alpha)^2}, \quad \alpha > \frac{1}{2}. \end{aligned}$$

Similarly,

$$\begin{aligned} \int_0^\infty f_2(y)^2 dy &= \frac{1}{B(\theta, \beta)^2} \int_0^\infty y^{2\theta-2} (1 + y)^{-2(\theta+\beta)} dy \\ &= \frac{B(2\theta - 1, 2\beta + 1)}{B(\theta, \beta)^2}, \quad \theta > \frac{1}{2}. \end{aligned}$$

The cross-product term is

$$\begin{aligned} \int_0^\infty f_1(y)f_2(y) dy &= \frac{1}{\beta^\alpha\Gamma(\alpha)B(\theta, \beta)} \int_0^\infty y^{\alpha+\theta-2} e^{-y/\beta} (1 + y)^{-(\theta+\beta)} dy \\ &= \frac{\Gamma(\alpha + \theta - 1)}{\beta^\alpha\Gamma(\alpha)B(\theta, \beta)} U\left(\alpha + \theta - 1, \alpha - \beta, \frac{1}{\beta}\right), \quad \alpha + \theta > 1. \end{aligned}$$

Substituting the above results, the Renyi entropy of order $r = 2$ is

$$\begin{aligned} H_2^{\text{Renyi}}(Y) &= -\ln \left[w^2 \frac{\Gamma(2\alpha - 1)}{2^{2\alpha-1}\beta\Gamma(\alpha)^2} \right. \\ &\quad \left. + (1 - w)^2 \frac{B(2\theta - 1, 2\beta + 1)}{B(\theta, \beta)^2} \right. \\ &\quad \left. + 2w(1 - w) \frac{\Gamma(\alpha + \theta - 1)}{\beta^\alpha\Gamma(\alpha)B(\theta, \beta)} U\left(\alpha + \theta - 1, \alpha - \beta, \frac{1}{\beta}\right) \right]. \end{aligned} \quad (46)$$

The Renyi entropy of order $r = 2$ exists provided $\alpha > \frac{1}{2}$ and $\theta > \frac{1}{2}$.

Similarly, closed-form expressions for the Renyi entropy can be derived for higher integer orders $r > 2$ using the same expansion technique.

10.1.2 Tsallis entropy

The Tsallis entropy of order $q > 0$, $q \neq 1$ is defined as

$$H_q^{\text{Tsallis}}(Y) = \frac{1}{q-1} \left[1 - \int_0^\infty f(y)^q dy \right].$$

Substituting the mixture PDF:

$$H_q^{\text{Tsallis}}(Y) = \frac{1}{q-1} \left[1 - \int_0^\infty (wf_1(y) + (1-w)f_2(y))^q dy \right].$$

Using the multinomial expansion (integer q):

$$(wf_1 + (1-w)f_2)^q = \sum_{k=0}^q \binom{q}{k} w^k (1-w)^{q-k} f_1(y)^k f_2(y)^{q-k},$$

So that

$$H_q^{\text{Tsallis}}(Y) = \frac{1}{q-1} \left[1 - \int_0^\infty \sum_{k=0}^q \binom{q}{k} w^k (1-w)^{q-k} f_1(y)^k f_2(y)^{q-k} dy \right] \quad (47)$$

For non-integer q , numerical integration is again required, subject to the existence of $\int_0^\infty f(y)^q dy$.

10.1.3 Tsallis entropy of order $q = 2$

For the special case $q = 2$, the Tsallis entropy reduces to

$$H_2^{\text{Tsallis}}(Y) = 1 - \int_0^\infty f(y)^2 dy.$$

Using the mixture representation $f(y) = wf_1(y) + (1-w)f_2(y)$, we have

$$f(y)^2 = w^2 f_1(y)^2 + (1-w)^2 f_2(y)^2 + 2w(1-w)f_1(y)f_2(y),$$

so that

$$\int_0^\infty f(y)^2 dy = w^2 \int_0^\infty f_1(y)^2 dy + (1-w)^2 \int_0^\infty f_2(y)^2 dy + 2w(1-w) \int_0^\infty f_1(y)f_2(y) dy.$$

Table 9 Comparison of Rényi entropy

Sr.No	Order	Gamma	Beta Prime	GBPD
1	r=0.5	1.837877	3.060271	2.143980
2	r= 1.75	1.418305	1.541828	1.930799
3	r=2	1.386294	1.475907	1.433111

Table 10 Comparison of Tsallis Entropy

Sr.No	Order	Gamma	Beta Prime	GBPD
1	q = 1.50	1.0351983	1.1140759	0.7614344
2	q = 1.75	0.8731115	0.9138322	0.7614344
3	q = 2	0.7500000	0.77142867	0.7614344

$$\begin{aligned}\int_0^\infty f_1(y)^2 dy &= \frac{\Gamma(2\alpha - 1)}{2^{2\alpha-1}\beta\Gamma(\alpha)^2}, & \alpha > \frac{1}{2}, \\ \int_0^\infty f_2(y)^2 dy &= \frac{B(2\theta - 1, 2\beta + 1)}{B(\theta, \beta)^2}, & \theta > \frac{1}{2}, \\ \int_0^\infty f_1(y)f_2(y) dy &= \frac{\Gamma(\alpha + \theta - 1)}{\beta^\alpha\Gamma(\alpha)B(\theta, \beta)}U\left(\alpha + \theta - 1, \alpha - \beta, \frac{1}{\beta}\right), & \alpha + \theta > 1.\end{aligned}$$

Hence, the Tsallis entropy of order $q = 2$ for the GBPD is

$$\begin{aligned}H_2^{\text{Tsallis}}(Y) &= 1 - \left[w^2 \frac{\Gamma(2\alpha - 1)}{2^{2\alpha-1}\beta\Gamma(\alpha)^2} \right. \\ &\quad + (1 - w)^2 \frac{B(2\theta - 1, 2\beta + 1)}{B(\theta, \beta)^2} \\ &\quad \left. + 2w(1 - w) \frac{\Gamma(\alpha + \theta - 1)}{\beta^\alpha\Gamma(\alpha)B(\theta, \beta)}U\left(\alpha + \theta - 1, \alpha - \beta, \frac{1}{\beta}\right) \right].\end{aligned}\quad (48)$$

The Tsallis entropy of order $q = 2$ exists provided $\alpha > \frac{1}{2}$ and $\theta > \frac{1}{2}$.

Similarly, closed-form expressions for higher integer orders $q > 2$ can be derived using the same expansion technique.

10.1.4 Special case: Shannon entropy

Both Rényi ($\alpha \rightarrow 1$) and Tsallis ($q \rightarrow 1$) reduce to Shannon entropy:

$$H(Y) = - \int_0^\infty f(y) \log f(y) dy.$$

Tables 9 and 10 show Reyni and Tsallis entropy values computed for gamma, beta prime, and the GBPD for different orders using numerical integration, highlighting uncertainty and non-extensive behavior across tail and body-dominated regimes. As order increases toward higher values, placing greater emphasis on the central part of the distribution, the entropy values of all models decrease and become closer, with the mixture maintaining stable values comparable to the competing distributions. For the heavy-tailed mixture distribution, Tsallis

entropy was evaluated for higher order ($q > 1$), ensuring existence and numerical stability of the entropy integral. This stability across different order values highlights the flexibility and robustness of the proposed GBPD distribution in capturing both tail and body characteristics of the data. To complement these numerical summaries, contour plots in Fig. 9 are employed to visualize the variation of entropy across order and parameter ranges.

11 Application

The mixture distribution of gamma and beta prime is suitable when data exhibit complex tail behavior, skewness, or heterogeneity that cannot be well modeled by simpler distributions. It's ideal in settings requiring detailed tail risk assessment, heavy-tailed phenomena, or complex biological, financial, or environmental data. The GBPD was compared to several standard distributions, including the normal, gamma, log-normal, and weibull distributions, to evaluate its suitability for modeling the data. Model comparison is conducted using AIC, BIC, CAIC, and HQIC, while goodness-of-fit is assessed via the Cramér–von Mises and Anderson–Darling statistics (Akaike 1974; Schwarz 1978; Hannan and Quinn 1979; Anderson and Darling 1952).

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with parameter vector $\theta \in \mathbb{R}^k$. Let $\ell(\theta)$ denote the log-likelihood function and $\hat{\theta}$ its maximum likelihood estimator. Further, let $F(x; \hat{\theta})$ be the fitted cumulative distribution function (CDF) and $F_n(x)$ the empirical CDF, where

$$U_{(i)} = F(X_{(i)}; \hat{\theta}), \quad i = 1, 2, \dots, n,$$

and $X_{(i)}$ denotes the i th order statistic.

Akaike Information Criterion (AIC)

$$\text{AIC} = -2 \ell(\hat{\theta}) + 2k$$

Consistent Akaike Information Criterion (CAIC)

$$\text{CAIC} = -2 \ell(\hat{\theta}) + k\{\ln(n) + 1\}$$

Hannan–Quinn Information Criterion (HQIC)

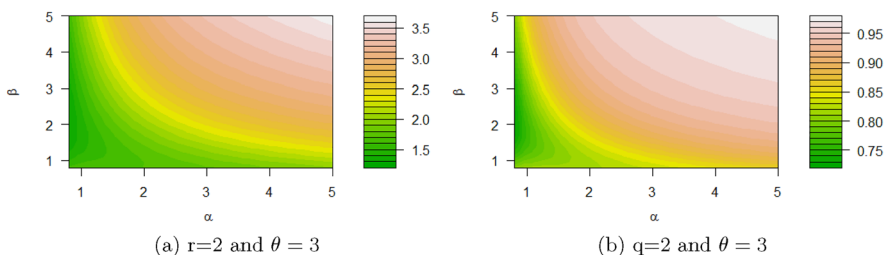


Fig. 9 Contoure plot for Renyi and Tsallis entropy

$$\text{HQIC} = -2\ell(\hat{\theta}) + 2k \ln(\ln n)$$

Bayesian Information Criterion (BIC)

$$\text{BIC} = -2\ell(\hat{\theta}) + k \ln(n)$$

Cramér–von Mises Statistic (CvM)

$$W^2 = \frac{1}{12n} + \sum_{i=1}^n \left(U_{(i)} - \frac{2i-1}{2n} \right)^2$$

Anderson–Darling Statistic (AD)

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i-1) [\ln U_{(i)} + \ln(1 - U_{(n+1-i)})]$$

Data Set I: The first positive sked data represents an uncensored data corresponding to remission times (in months) of a random sample of 128 bladder cancer patients Lee and Wang (n.d.); Merovci (2013).

0.08, 2.09, 3.48, 4.87, 6.94, 8.66, 13.11, 23.63, 0.20, 2.23, 3.52, 4.98, 6.97, 9.02, 13.29, 0.40, 2.26, 3.57, 5.06, 7.09, 9.22, 13.80, 25.74, 0.50, 2.46, 3.64, 5.09, 7.26, 9.47, 14.24, 25.82, 0.51, 2.54, 3.70, 5.17, 7.28, 9.74, 14.76, 26.31, 0.81, 2.62, 3.82, 5.32, 7.32, 10.06, 14.77, 32.15, 2.64, 3.88, 5.32, 7.39, 10.34, 14.83, 34.26, 0.90, 2.69, 4.18, 5.34, 7.59, 10.66, 15.96, 36.66, 1.05, 2.69, 4.23, 5.41, 7.62, 10.75, 16.62, 43.01, 1.19, 2.75, 4.26, 5.41, 7.63, 17.12, 46.12, 1.26, 2.83, 4.33, 7.66, 11.25, 17.14, 79.05, 1.35, 2.87, 5.62, 7.87, 11.64, 17.36, 1.40, 3.02, 4.34, 5.71, 7.93, 11.98, 18.10, 1.46, 4.40, 5.85, 8.26, 11.98, 19.13, 1.76, 3.25, 4.50, 6.25, 8.37, 12.02, 2.02, 3.31, 4.51, 6.54, 8.53, 12.03, 20.28, 2.02, 3.36, 6.76, 12.07, 21.73, 2.07, 3.36, 6.93, 8.65, 12.63, 22.69, 5.49.

Table 11 presents the model selection criteria for the competing distributions, including AIC, CAIC, BIC, HQIC, and AD, allowing a comparison of model fit and complexity. Lower values indicate a better balance between goodness-of-fit and model parsimony. Figures 10 and 11 presents the PDF plots of various distributions along with the QQ plot of the GBPD model.

Data Set II The second dataset is characterized by a positively skewed distribution with a long right tail. It comprises 46 observations representing the active repair times (in hours)

Table 11 Log-likelihood estimated value and respective AIC, CAIC, HQIC, BIC, CvM and AD for different distributions of Data set I

Sr.No	Distribution	−2log L	AIC	CAIC	HQIC	BIC	CVM	AD
1	GBPD	820.66	826.66	841.22	830.14	835.22	0.03	0.17
2	Normal	964.41	968.41	978.12	970.73	974.12	1.76	9.82
3	Gamma	826.78	830.78	840.48	833.10	836.48	0.14	0.77
4	LogNormal	830.24	834.24	843.94	836.56	839.94	0.12	0.80
5	Weibull	828.21	832.21	841.92	834.52	837.92	0.15	0.96
6	Beta prime	850.83	854.83	864.53	857.14	860.53	0.47	2.71

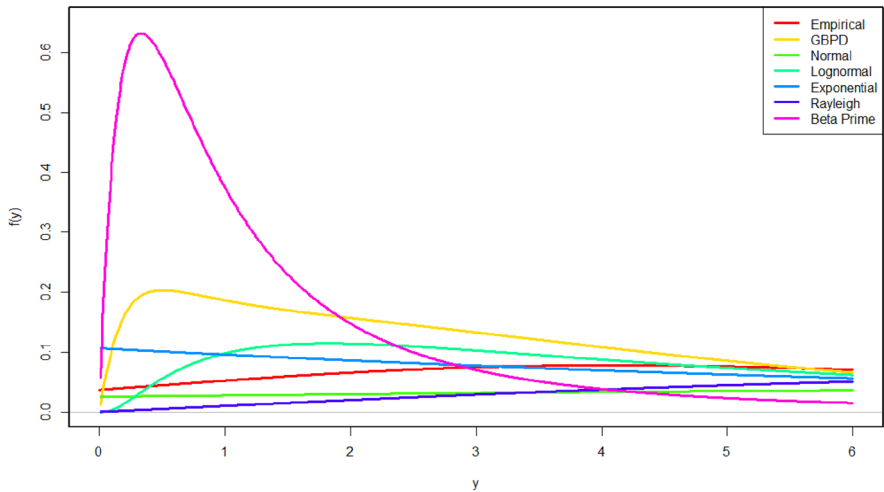


Fig. 10 Pdf plot for data set I

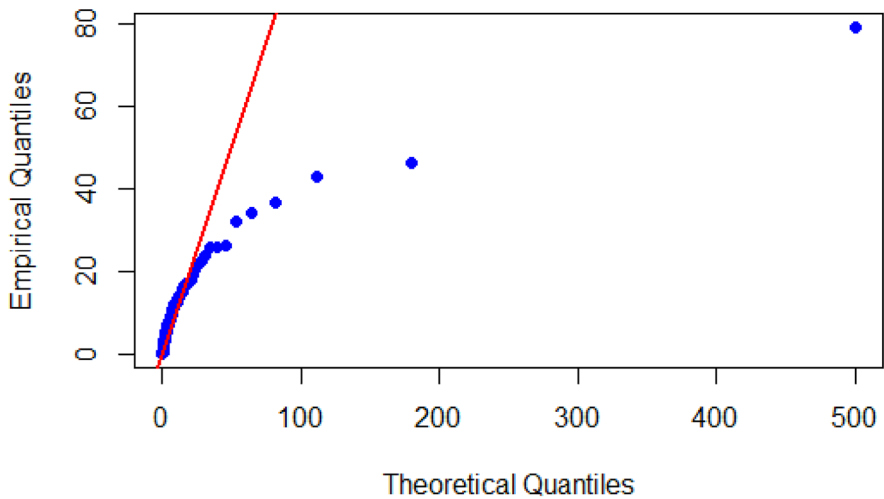


Fig. 11 QQ plot for data set I

of an airborne communication transceiver. The dataset was first published by William H. Von Alven (1964) in a landmark investigation of military system reliability (Von Alven 1964) and has since been widely used in reliability and lifetime data analysis. In statistical literature, it is often cited via its prominent re-analyses by Chhikara and Folks (Chhikara and Folks 1989).

0.2, 0.3, 0.5, 0.5, 0.5, 0.6, 0.6, 0.7, 0.7, 0.7, 0.8, 0.8, 1.0, 1.0, 1.0, 1.0, 1.1, 1.3, 1.5, 1.5, 1.5, 1.5, 2.0, 2.0, 2.2, 2.5, 2.7, 3.0, 3.0, 3.3, 4.0, 4.0, 4.5, 4.7, 5.0, 5.4, 5.4, 7.0, 7.5, 8.8, 9.0, 10.3, 22.0, 24.5, 25.0, 28.8

We utilized graphical

Table 12 Log-likelihood estimated value and respective AIC, CAIC, HQIC, BIC, CvM and AD for different distributions of Data set II

Sr.No	Distribution	$-2\log L$	AIC	CAIC	HQIC	BIC	CvM	AD
1	GBPD	218.09	226.09	237.40	228.83	233.40	0.0338	0.2462
2	Normal	306.94	310.94	316.60	312.31	314.60	1.1375	6.3766
3	Gamma	232.35	236.35	242.01	237.72	240.01	0.2590	1.6253
4	LogNormal	220.42	224.42	230.08	225.79	228.08	0.0584	0.4152
5	Weibull	230.14	234.14	239.79	235.51	237.79	0.1637	1.1838
6	Beta prime	220.32	224.32	229.98	225.69	227.98	0.0527	0.3414

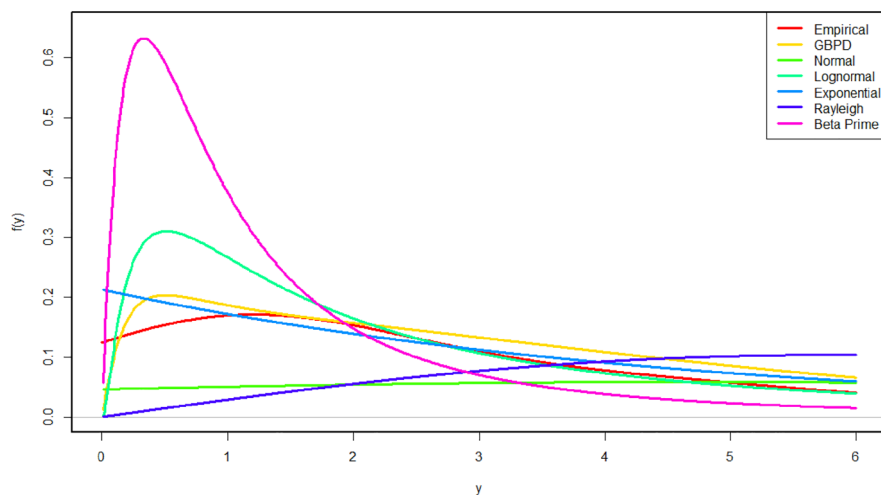
**Fig. 12** Pdf plot for data set II

Table 12 presents a comparative assessment of the GBPD mixture model against five competing lifetime distributions using the Active Repair Times dataset. The results demonstrate that the proposed GBPD model provides the best overall fit among the considered models, achieving the smallest negative log-likelihood value 218.09. Furthermore, the GBPD model attains the lowest goodness-of-fit statistics, with Cramér–von Mises ((CvM = 0.0338)) and Anderson–Darling ((AD = 0.2462)) values considerably smaller than those of the competing models. These results indicate a superior agreement between the fitted and empirical distributions across the entire range of observations. Although information criteria such as AIC and BIC slightly favor some simpler two-parameter models due to their lower complexity penalties, the substantial reduction in the CvM and AD statistics achieved by the GBPD model justifies the inclusion of additional parameters. The enhanced flexibility of the mixture framework enables it to capture both the pronounced skewness present in the main body of the data and the behavior of extreme repair-time observations more effectively than the competing distributions. Figures 12 and 13 displays the pdfs of the competing distributions and the QQ plot for the GBPD, illustrating how well the proposed model aligns with the empirical data.

Data set III-The third dataset comprises 84 observations representing the failure times of aircraft windshields. Although the distribution is approximately symmetric and close to

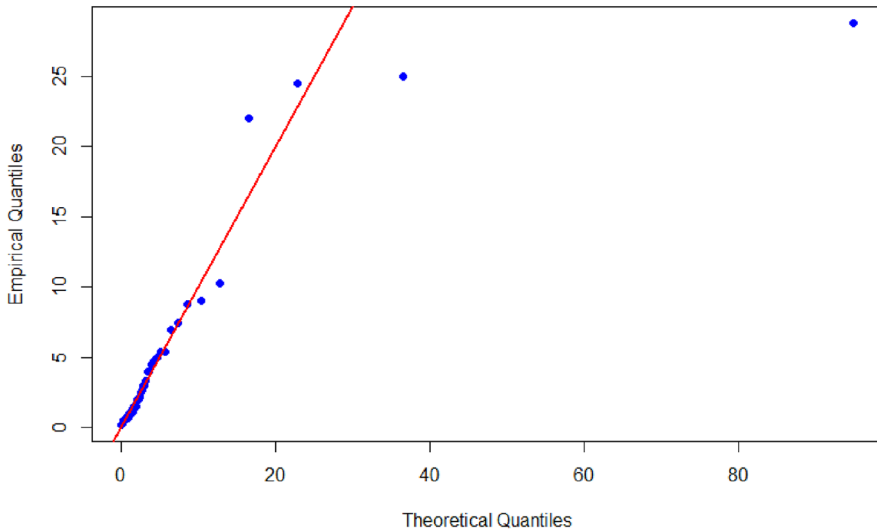


Fig. 13 QQ plot for data set II

Table 13 Log-likelihood estimated value and respective AIC, CAIC, HQIC, BIC, CvM and AD for different distributions of Data set III

Sr.No	Distribution	$-2\log L$	AIC	CAIC	HQIC	BIC	CvM	AD
1	GBPD	256.20	264.20	277.92	268.11	273.92	0.0706	0.5450
2	Normal	256.24	260.24	267.11	262.20	265.11	0.0911	0.6021
3	Gamma	273.87	277.87	284.74	279.83	282.74	0.1715	1.3788
4	LogNormal	307.84	311.84	318.70	313.80	316.70	0.05508	3.7077
5	Weibull	260.11	264.11	270.97	266.06	268.97	0.0513	0.5606
6	Beta prime	310.60	314.60	321.46	316.55	319.46	0.06619	4.2046

normal, it exhibits a mild negative (left) skewness. Owing to its practical significance and well-documented characteristics, this dataset has become a standard benchmark for assessing reliability models and lifetime distributions in the statistical literature. The data tracks the structural lifespan and failure rates of airplane windshields. The unit of measurement for these failure times is typically recorded in thousands of hours Tahir et al. (2015).

0.040, 1.866, 2.385, 3.443, 0.301, 1.876, 2.481, 3.467, 0.309, 1.899, 2.610, 3.478, 0.557, 1.911, 2.625, 3.578, 0.943, 1.912, 2.632, 3.595, 1.070, 1.914, 2.646, 3.699, 1.124, 1.981, 2.661, 3.779, 1.248, 2.010, 2.688, 3.924, 1.281, 2.038, 2.823, 4.035, 1.281, 2.085, 2.890, 4.121, 1.303, 2.089, 2.902, 4.167, 1.432, 2.097, 2.934, 4.240, 1.480, 2.135, 2.962, 4.255, 1.505, 2.154, 2.964, 4.278, 1.506, 2.190, 3.000, 4.305, 1.568, 2.194, 3.103, 4.376, 1.615, 2.223, 3.114, 4.449, 1.619, 2.224, 3.117, 4.485, 1.652, 2.229, 3.166, 4.570, 1.652, 2.300, 3.344, 4.602, 1.757, 2.324, 3.376, 4.663.

The goodness-of-fit measures for the fitted distributions are presented in Table 13. The proposed Custom Mixture distribution yielded the smallest negative log-likelihood value ($-2\log L = 256.23$) and the lowest Anderson–Darling statistic ($AD = 0.5450$) among all competing models, indicating a superior overall fit to the data, particularly in the tail

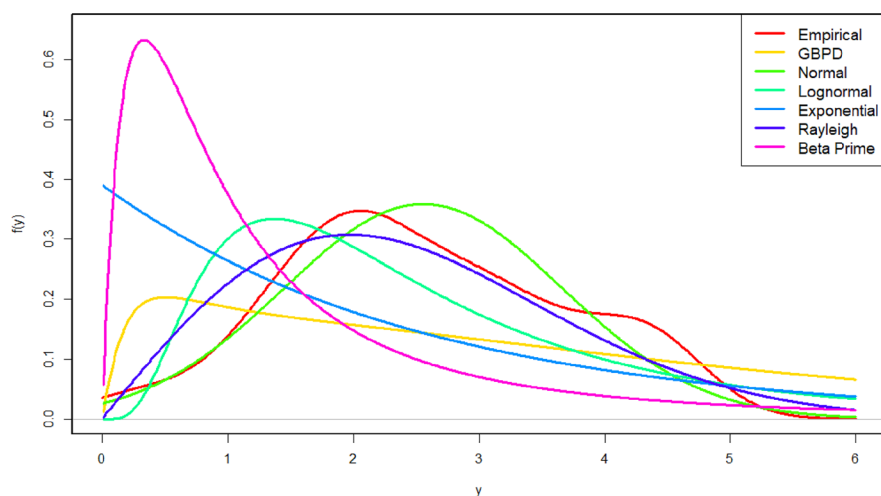


Fig. 14 Pdf plot for data set III

regions. The model also produced a relatively small Cramér–von Mises statistic, further supporting its adequacy.

Although the Normal distribution attained slightly smaller values of AIC and BIC, these criteria impose a stronger penalty for additional parameters. Since the Custom Mixture model contains one extra parameter compared with the classical two-parameter distributions, its information criteria are marginally larger despite achieving a better likelihood fit. Moreover, the substantially lower Anderson–Darling statistic suggests that the proposed model provides a more accurate representation of the distributional behavior, especially

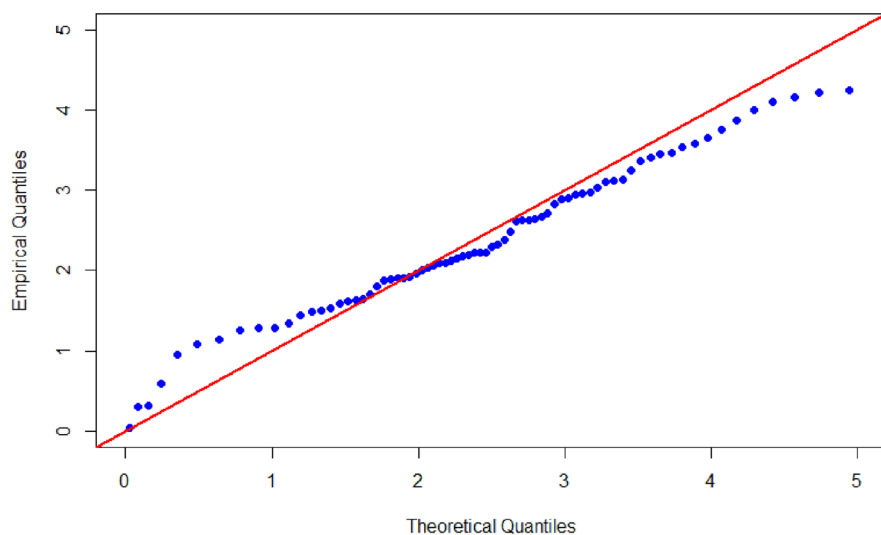


Fig. 15 QQ plot for data set III

in the tails. Therefore, considering both likelihood-based and goodness-of-fit measures, the Custom Mixture distribution offers a competitive and reliable model for describing the aircraft windshield failure time data. Figures 14 and 15 displays the pdfs of the competing distributions and the QQ plot for the GBPD, illustrating how well the proposed model aligns with the empirical data.

The three datasets considered in this study represent distinct distributional characteristics, namely a moderately skewed dataset, a positively skewed dataset with a pronounced right tail, and an approximately symmetric dataset that is close to normal. The goodness-of-fit results demonstrate that the proposed model provides an excellent fit across all three scenarios. In particular, the proposed model exhibits its strongest performance for the positively skewed dataset, where it achieves the best overall fit among all competing models.

12 Conclusion

This paper introduced a flexible mixture distribution obtained by combining the gamma and beta-prime distributions with data-driven mixing weights. The proposed model effectively captures both the central behavior and tail characteristics of the data, thereby overcoming the limitations of traditional single-component distributions. Parameter estimation was carried out using maximum likelihood estimation, and the performance of the proposed distribution was evaluated through goodness-of-fit measures and comparative analysis with well-known competing models.

To further investigate distributional uncertainty and complexity, Renyi and Tsallis entropies were evaluated for different values of the order parameter. The entropy analysis showed that the proposed mixture achieves balanced uncertainty, lying between the light-tailed gamma and heavy-tailed beta-prime distributions for tail-sensitive orders, while exhibiting stable behavior for higher orders that emphasize the body of the distribution. In addition, inequality and dispersion characteristics were examined using the mean deviation and the Gini index, which provide robust, moment-free measures of spread and concentration. These results collectively highlight the adaptability, robustness, and interpretability of the proposed GBPD across varying uncertainty and dispersion regimes.

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Data availability All data supporting the findings of this study are fully available in the article and are clearly cited.

Code availability The code supporting the findings of this study is available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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